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From Video to Voice: How YouTube Shapes Public Discourse on Income Inequality

Author: Linying Li

Student ID Nr.: 22-738-702

Supervisor: Prof. Dr. Fabrizio Gilardi

Department of Political Science

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Abbreviations

Abbreviations	Definitions
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
COP	Conference of Parties
COVID-19	Coronavirus disease
CSV	Comma Separated Values
e.g.	For example
E-I index	External-internal Index
et al.	And others
GPU	Graphics Processing Unit
GB	Gigabyte
i.e.	In other words
LDA	Latent Dirichlet Allocation
LSA	Latent Semantic Analysis
MLM	Masked Language Model
NLP	Natural Language Processing
NSP	Next Sentence Prediction
RNN	Recurrent Neural Network
STM	Structural Topic Modeling
TF-IDF	Term Frequency-Inverse Document Frequency

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Abstract

Income inequality is a critical issue with widespread concern, yet the discussion on major online video platforms like YouTube remains under-explored. This study investigates YouTube's unique characteristics and their impact on public discourse regarding income inequality, focusing on content production, distribution, and consumption. Analysis reveals four distinct topics: Mainstream Narrative, Wealth Accumulation, Broken Capitalist System, and Bad Government Intervention. Findings indicate heterogeneous political interactions on YouTube, with minimal evidence of echo chambers. The complexity of YouTube's interaction environment and its potential to foster diverse discussions are highlighted. Results emphasize the importance of platform design in shaping public discourse and suggest further investigation into the role of recommendation algorithms and other platform-specific features in influencing user engagement and political polarization.

1. Introduction

About 60 years ago, Jürgen Habermas introduced the concept of the public sphere, rooted in the ancient Greek tradition and revitalized in the 18th-century bourgeois salons. The public sphere emerged as an intermediate field between public power and individuals during the transition from feudalism to a capitalist society. As the bourgeoisie demanded participation in state governance, the public sphere assumed a political function. According to Habermas, the public sphere provides an open arena for individuals to discuss social issues with public authorities. Initially, newspapers and magazines carried the political public sphere. However, 20th-century mediums like television and radio, despite erasing the distance between the audience and the publication, required significant capital, allowing market forces to encroach on the public sphere (Habermas 2006).

In the 21st century, social media has become a pivotal conduit for information dissemination and public discourse, transforming societal communication paradigms. According to Pew Research (2023), about 30% of American adults regularly get their news from Facebook and 26% from YouTube. Unlike traditional media, social media has inherent social interaction features. The digitalization of public communication offers unlimited point-to-point propagation networks, enabling every user to create content (Bruns and Highfield 2015). This decentralization empowers citizens to voice their opinions and participate in democracy. The influence of social media can be measured directly through clicks and shares, prompting traditional media to adapt to these platforms' rules and produce more emotional and unique posts (Fischer and Jarren 2023). However, the absence of gatekeepers leads to the proliferation of irresponsible information. Habermas (2023) recently noted that boundless communication networks can spread centrifugally while condensing into communication circuits that dogmatically seal off each other, counteracting the integrating power of traditional media.

In recent decades, income inequality and political polarization have increased dramatically in the United States and Western Europe. Literature suggests that income inequality drives political polarization. Studies (Voorheis et al. 2015; McCarty et al. 2006) found that party policies tend to polarize to appeal to voters as inequality rises. Stewart et al. (2020) argue that economic insecurity leads individuals to interact more with in-group members, fostering in-group polarization. When disadvantaged groups' interests go unnoticed, their "Say No" power can become anarchy (Habermas 2023). Empirical research (Engler and Weisstanner 2021; Stoetzer et al. 2023)

confirmed that rising economic inequality has driven the rise of populist and radical political parties. Income inequality has become an important topic of discussion in the political arena.

Social media has provided a platform for egalitarian public discourse and increased polarization. Sunstein (2018) argues that social media makes it easier for people with similar views to find each other and exclude opposing views, which threatens democracy by fragmenting public discourse. Empirical evidence shows that social media users tend to interact within groups that are politically aligned, both in terms of following networks (Barberá et al. 2015; Rivero 2017; Garimella and Weber 2017) and retweeting or resharing patterns (Conover et al. 2011; Barberá 2015; Bakshy et al. 2015).

Despite numerous studies on social media and political polarization, the extent to which existing empirical research addresses theorists' specific concerns remains questionable. Habermas (2006) worries that social media fragmentation prevents the public sphere from providing stable political input to democracy. Current research often simplifies social media content into two political labels, left-wing and right-wing, overlooking the depth of content analysis. For instance, narratives about income inequality can be broadly categorized as left-wing or right-wing, but they vary significantly in their understanding and interpretation of inequality. Additionally, within the same political stance, the intensity varies, with some discourses strongly advocating for redistribution while others are more moderate. How are these narrative differences presented on social media? How are political views linked to users' media consumption? This thesis aims to fill these gaps by exploring the narrative divergence and polarization on the most popular video platform, YouTube, offering a more nuanced analysis of social media discussions on income inequality. This thesis focuses on a critical question: How does social media shape the discussion of income inequality?

To answer the question, this thesis collected 245 YouTube videos related to income inequality and the 341,556 comments belonging to them via the YouTube API. The empirical analysis is conducted on three dimensions of social media: content production, distribution, and consumption (Klinger and Svensson 2015). Firstly, the research explores the narrative divergence about income inequality on YouTube using BERTopic, a topic modeling method. Given the decentralized nature of social media and the lack of gatekeepers, different narratives may emerge on the same topic. Secondly, using the fine-tuned BERT model to classify comments, this paper measures the polarization of interaction networks using the Krackhardt E-I index. Thirdly, the impact of video content on viewer comments was measured using the Manifesto model and a linear regression

model. This part examines to what extent the video's relevance to a particular viewpoint attracts comments that echo the same position.

The thesis is organized as follows: The literature review chapter reviews and summarizes the concepts related to this study and the process of developing empirical research. This chapter concludes with a summary of the current state of research and identifies research gaps. The Data chapter describes in detail the data collection process, which is divided into two parts: the collection of video data and the collection of comment data. Additionally, the Data chapter describes the collected data. The Theoretical Framework chapter addresses the research questions by incorporating YouTube's unique characteristics and identifying gaps in current academic research. Based on existing studies, this chapter formulates hypotheses to guide the empirical analysis. Chapters five, six, and seven present the empirical analyses detailing the methodologies, results, and interpretations of the limitations conducted. Finally, the conclusion reviews the main findings of the thesis, its contribution to academic research, and points to future research directions.

This research contributes to understanding social media discussions on the critical issue of income inequality. It analyzes whether social media serves as a dividing or unifying force in public discourse on income inequality. The findings are essential for evaluating the role of digital media in fostering a common understanding of key social issues and for developing strategies to enhance constructive public debate.

2. Literature Review

The literature review systematically examines existing research, focusing on conceptual development and the evolution of empirical studies. It is structured into four sections. The first section explores the definition of social media and its impact on the public sphere and opinion formation in democratic societies. This part addresses fundamental questions about what social media is and how it influences public discourse and democracy. The second section delves into the characteristics of social media, addressing concepts such as information bubbles and echo chambers. It examines how these features shape the consumption and sharing of information. The third section investigates the social consequences of social media, particularly its role in societal polarization. This part reviews empirical studies that demonstrate how social media contributes to polarization. The fourth section summarizes the various narratives surrounding income inequality, as discussed in the existing literature. It provides a theoretical foundation for the explanations presented in the subsequent empirical chapter. The conclusion reviews the main ideas and findings in the literature, identifying existing research gaps and providing a basis for the research questions and hypotheses.

2.1. Social Media and the Transformation of the public sphere

As Habermas (1991) described a structural transformation in Western society in the last century, he conceived a simple and efficient space that coordinates elites and masses. Since television and radio boosted the spread and influence of public discourse, the population was no longer directly involved in debates and discussions (Bruns and Highfield 2015). In this space, a few media organizations with licenses offer a virtual arena shared by the community. The arena is populated by elite actors with influence, while the public acts as the audience, passively watching the events on the stage (Habermas 2006: 416). While Habermas envisioned a public sphere still centered on the elite, the transformation of the public sphere from café to mass media has not ended.

The appearance of social media reshaped the public sphere. Social media consists of various web and mobile platforms facilitating interaction among individuals and groups, including microblogging services like Twitter, social networks like Facebook, streaming media like YouTube, and social news forums like Reddit (Nahon 2015). Each platform has its own specific features and guidelines, which shape distinct power dynamics among users (Nahon 2015). Klinger and Svensson

(2015) refer to social media as network media. They argue that social media differ in three dimensions, production, distribution, and consumption, compared to traditional media. The purpose of information production on social media is to generate personalized content, and news is published in mass media because the staff thinks the audience should know about it. Thus, the production logic on social media is personalization and attention maximization rather than newsworthiness or public interest. Traditional media, such as print and cable television, rely heavily on a relatively fixed subscriber base. In contrast, content distribution on social media platforms is predominantly driven by users. This means that content becomes popular not necessarily because of its factual accuracy or depth but because it resonates with the beliefs or interests of a particular group and is pushed to visibility through network effects. Traditional media involves a passive consumption model, and users and audiences receive it in a relatively passive manner. In contrast, social media platforms facilitate an interactive model of media use. Social media allows users to discern, combine, add, or delete information according to their preferences. Users not only consume but also actively contribute to content creation and dissemination.

Since social media has become a source of news (Pew Research Center 2012), the traditional single media environment where dominated by a few major organizations, has transformed into a “diverse, complex and even confused media ecology” (Bruns and Highfield 2015: 57). This shift has made the public sphere much more accessible and interactive, enabling not just elite actors but also ordinary citizens to contribute to and shape public discourse. Bruns and Highfield (2015) introduce the concept of “produsage” to describe this new model where the distinctions between producers and users of content become increasingly blurred, fostering a participatory political culture in the digital age. This evolution also poses challenges to the traditional triangle of political communication, where the supply side (politicians and journalists) once determined the discourse. The focus has now moved toward the demand side, with politicians and journalists adapting to the interests and behaviors of voters (Broersma and Graham 2015). In this sense, social media has contributed to the democratization of political communication (Benkler 2006: 272). Thompson (2005) argued that technical advances have changed the essence of communication. Digital technologies enable information to be shared instantaneously across the globe. Unlike traditional forms of media that were limited in their reach, digital platforms have a global audience. Mediated visibility in the digital age plays a significant role in shaping public narratives. The ability to

disseminate information widely and rapidly through digital platforms empowers individuals and grassroots movements to challenge traditional power structures.

However, the consequence of this adaptation is that sub-domains divided by different interests replaced the original unifying public attention. The alternative explanation tries to segment the public sphere into a number of subsets. Dahlgren (2005) argues that an integrated public sphere has disappeared in social fact, and spaces must exist in which diverse groups find support and expression. Habermas (2006) provided the term “issue public,” which describes the essence of this new, fragmented landscape of public discourse. An “issue public” refers to a segment of the broader public that organizes itself around specific topics or concerns, distinguishing itself from the mass public by its focused interest and often more active engagement. Some people care passionately about growing inequality, while others may care little about inequality but a great deal about climate change. Schmidt (2014) further outlined the concept of “personal public” on social media. Compared with the traditional public, the personal public on social media, e.g., Twitter, has three features: personalized information, clear networked audience, and conversation instead of publishing.

Easton (1965) describes political systems as mechanisms that continuously process inputs, such as demand and support, and produce outputs, like policies and decisions. For democracy, public opinion is the most crucial input. Public opinion reflects the public’s expectations, needs, and preferences and is a factor that must be considered when formulating government policies in a democracy. It significantly influences government policy-making, as a government’s legitimacy and effectiveness partly depend on how responsive it is to public opinion. Habermas (1996) describes opinion formation in democratic societies as occurring within a flexible and inclusive network of overlapping subcultural groups, which have dynamic social compositions and loosely defined topics. However, whether social media effectively facilitates shared opinion formation across these diverse groups is debatable. The debate centers around whether social media platforms truly foster a collective discourse conducive to democratic deliberation or whether they amplify division by creating echo chambers where like-minded individuals reinforce each other’s beliefs. This phenomenon could potentially undermine the public sphere’s integrative functions as Habermas envisioned, where a diverse population engages in rational debate to reach a consensus or at least mutual understanding.

In summary, social media has fundamentally reshaped the public sphere by changing how information is produced, distributed, and consumed. Unlike traditional media, where professionals curate content and audiences consume it passively, social media platforms empower users to actively create and share content. This shift has democratized political communication, allowing ordinary citizens to participate in public discourse alongside elites. However, it has also fragmented the public sphere, with people engaging around specific interests rather than a unified public agenda. This fragmentation raises critical questions about the effectiveness of social media in fostering collective discourse and shared opinion formation, both of which are crucial for the health and function of democratic deliberation. In the following sections, these issues will be examined in detail, focusing on how selective exposure to information and the resulting political polarization impact the integrity of the public sphere.

2.2. The echo chamber on social media

Before 2300 years ago, Aristoteles (1944: §1286a21), in arguing for democracy, believed that the multitude is more judgmental than a single person since they can't be wrong at the same time. This view is based on the fact that discussion exchanges knowledge, provides diverse arguments, and avoids biased deductions. However, from the psychological perspective, human nature seeks information that aligns with their existing views to reduce psychological discomfort caused by conflicting information (Sears and Freedman 1967). People prefer media that aligns with their beliefs and tend to avoid content that contradicts them (Festinger 1957). On the one hand, the rise of social media transformed how communication connects; it pushed more people to join the communication process, and on the other hand, it empowered individuals to choose what to see. Terms such as information cocoon, filter bubble, epitome bubble, and echo chamber were widely in the literature and academic discussion used and discussed. This section will explore the definition, causes, and effects.

Before the emergence of social media, Negroponte (1995) found the role of digitalization in news selection and first exposed the concept of "The Daily Me." This term describes a future where news and media are so tailored to individual preferences that each person only encounters information reinforcing their existing beliefs, biases, and preferences.

Pariser (2011) expanded on these concerns in his seminal book, which first brought attention to the phenomenon of filter bubbles. He worried that the algorithm's personalized sorting would create

a unique information environment for each individual, which would lead to the intellectual isolation of internet users. For example, if a user always views left-wing posts, then the algorithm displays more left-wing posts based on the user's behavior. Thereafter, the shared information disappears among different users. The Internet ostensibly offers the potential to democratize access to information and distribute authority widely. However, in practical terms, it exerts unprecedented centralized control over what people consume and the opportunities available to them.

Sunstein (2018) introduced the concept of the echo chamber, an online space where individuals are surrounded by and continuously exposed to their own views, leading to a reinforcement of their beliefs. He argued that social media facilitates this isolation from opposing viewpoints, which can undermine democratic debate. Echo chambers create a shared frame of reference and reinforce positive feedback loops among like-minded individuals (Jamieson and Capella 2008). In academic discussions, the terms "filter bubble" and "echo chamber" are often used interchangeably, but distinguishing between them is crucial. Nguyen (2018) clarified that while an echo chamber involves the active exclusion and discrediting of outside voices within a social group, a filter bubble refers to the personalized information that individuals receive online, curated by algorithms based on user preferences. Both phenomena result in the reinforcement of existing beliefs and limited exposure to diverse viewpoints. However, echo chambers are more about social interactions and group dynamics, whereas filter bubbles are driven by algorithmic content personalization.

Literature often links echo chambers to polarization (Vicario et al. 2016; Cinelli et al. 2021), showing that like-minded messages spread among people who share the same political stance. Sunstein (2018) connects echo chambers to group polarization, suggesting that polarization involves avoiding interaction with people of different political beliefs. He thinks the concept of polarization is avoiding interaction with people of different political beliefs. In addition, the overlap between echo chambers and polarization is closely linked to how social media works, including how content is created, shared, and consumed. An echo chamber occurs when groups are exposed to limited information, while polarization is the social outcome of these echo chambers. On platforms like Twitter, information spreads through retweets as users share content with their networks and create new distribution points. Therefore, echo chambers and group polarization are the same on Twitter.

Empirical researches confirm selective exposure to social media. Bakshy et al. (2015) utilized a dataset of shared content on Facebook to analyze the exposure to ideologically diverse news and opinions on the platform. They observed substantial isolation in the political contents shared by

users, with links that align with either conservative or liberal populations. An et al. (2013) conducted a detailed study that revealed that users on Facebook predominantly share and consume news that reinforces their existing beliefs, thereby avoiding information that contradicts their viewpoints. This pattern is especially pronounced among users with distinct political affiliations, who exhibit a significant preference for like-minded news sources. Mukerjee and Yang's study (2021) showed that users typically avoid news from sources they politically disagree with and prefer news from sources they agree with, regardless of how popular the news is on social media. Similarly, Mothes and Ohme (2019) found that people with strong political affiliations tend to stick to news that supports their views, especially during important events like elections. Cinelli et al. (2021) conducted research on four major social media platforms (Facebook, Twitter, Reddit, and Gab) and found that these platforms significantly contribute to the formation of echo chambers by promoting content that is highly aligned with users' existing beliefs. Vicario et al. (2016) focused their research on the Brexit debate on Facebook by examining how users interacted with content related to Brexit on the platform. Through their analysis, they identified the formation of two distinct communities or echo chambers within the online discussions. Cinelli et al. (2020) studied the interaction patterns of 14 million Facebook users with 583 news outlets over a six-year period. They utilized topic modeling algorithms, specifically Latent Dirichlet Allocation (LDA), to infer the latent topical structure of the collected news posts. Their findings revealed that users, regardless of their online activity levels or time spent online, tended to interact with a limited number of news outlets. Flaxman et al. (2016) collected web-browsing data using the Bing Toolbar, which provides a collection of websites opened in the browser for 1.2 million US-located users over a three-month period in 2013. They focused on 50,000 users who actively read the news. They found that articles discovered through social media or web search engines were associated with higher ideological segregation compared to those accessed by directly visiting news sites.

However, despite literature that exposes the existence of echo chambers on social media, the size of echo chambers receives questioning. Some scholars (Nelson et al. 2017; Messing and Westwood 2014) argue that echo chambers only occur in a minority of the population. For example, Flaxman et al. (2016) found that most online news consumption still involved individuals directly visiting the home pages of mainstream news outlets. Fetcher et al. (2021) used survey data and online tracking methods to identify echo chambers. They found that politically partisan online news echo chambers occur, but the size of the echo chamber varies between different countries. In Britain, the number of

people living in echo chambers is only 5%, but in the United States, it is 10%. Similarly, Dubois and Blank (2018) found that about 10% of people in Britain never see opposite political content on social media. Nelson et al. (2017) conducted a network analysis to examine audience duplication among different online political news. Their study found that online political news audiences congregated around a few popular news sites, both mainstream and partisan, which means that online political news audiences exhibit diversity and interconnectedness. Messing and Westwood (2012) focused on how social media endorsements influence news consumption habits online. Social endorsement means highly recommended content from other people. They found that the effect of social endorsements was particularly pronounced for individuals selecting articles from sources that were ideologically misaligned, which indicates that social cues can override partisan biases in news consumption. Their study suggests that social media platforms, by leveraging social endorsements, can increase users' exposure to a variety of news and politically diverse information. Furthermore, Bakshy et al. (2015) also found that individuals were exposed to cross-cutting content, particularly traditional media shared on social media. Selective exposure to social media also depends on an individual's demographic and socioeconomic background. Kumpel (2020) utilized data from the Reuters Institute Digital News Report 2018 to examine the inequalities in news exposure and engagement on social media platforms. He found that highly skilled and well-educated individuals were more likely to benefit from the opportunities offered in high-choice media environments. The research also suggested that incidental news exposure on social network sites could potentially contribute to closing information gaps by exposing users to news content through sharing activities of friends or inadvertent exposure to discussions about current events. Hameleers et al. (2018) employed survey experiments with a diverse sample of Dutch citizens. They found individual experience has a strong impact on populist media content exposure.

This section reviews the literature on selective information exposure in relation to social media. Taken together, these previous findings from selective information exposure research make it hard to craft a definitive statement on how extensive and harmful echo chambers really are. While some studies (Bakshy et al. 2015; An et al. 2013; Mukerjee and Yang 2021) have demonstrated a high degree of information and ideological homogenization, arguing that users are exposed to mostly content that echoes their views, other studies have shown that information exposure on social media is influenced by a variety of factors and thus the size of affected population is questionable (Nelson et al. 2017; Messing and Westwood 2014). This inconsistency highlights the need for further

research to understand how and when selective exposure occurs on social media and its broader implications. In the next section, the consequences of this phenomenon will be further discussed.

2.3. The political polarization and social media

According to the law of group polarization articulated by Sunstein (1999), when like-minded individuals engage in discussions within a group, their opinions tend to become more extreme compared to their initial individual beliefs. This phenomenon occurs because each group member is exposed to a reinforcing array of similar views that magnify their initial inclinations. In the context of social media, this process is further intensified. Platforms such as Facebook, Twitter, and YouTube provide not only a space for like-minded individuals to convene but also algorithmic mechanisms that filter and funnel content that is likely to be engaging. Before delving into the specific impact of social media on polarization, it is crucial to first understand the broader concept of polarization through a review of existing research. Establishing a clear definition of polarization is a fundamental initial step in this process.

Prinz (2021) defines political polarization as two groups perceiving political and confrontational borders between them. The political border includes not only partisan membership but also region, religion, class, and race. When these features have the ability to influence people's political behavior, they construct political borders. Political borders alone do not shape polarization. Polarization is shaped by groups of people with different political characteristics when they have completely opposite political choices. For instance, pro-abortion and anti-abortion people in the United States have completely different political affiliations. The existence of political borders inherently assimilates people's political opinions within a group. Opposing borders make for conceptual confrontation between different groups, and thus, support for the views of other political groups is intolerable. McCarty (2019) divides political polarization into policy polarization, ideological polarization, and partisan polarization. Policy polarization is defined as two viewpoints that are difficult to compromise on a particular policy. This means that people either completely accept a policy or completely reject it. Ideological polarization implies that society is sharply divided in its general orientation towards politics and governance. Most people are at opposite ends of the ideological spectrum. Compared with ideological polarization, partisan polarization is thought of as a rise in partisanship. Partisan polarization refers to any preferences and attitudes of individuals towards the political party they belong to. Baldassarri and Page (2021) distinguish

between ideological polarization and emotional polarization. While ideological polarization depends on the individual's position and interest, emotional polarization is not related to socio-demographics and is rooted in the group sense. In precise terms, it divides the political emotions of individuals into "in-group" loyalties and "out-group" animosities. With emotional polarization, Individuals' attitudes are highly correlated with partisan labels and even spill over into other non-political areas, with high identification with in-group members and hostility toward out-group members. For instance, Iyengar and Westwood (2015) found both Republicans and Democrats tend to view the opposing party negatively and the same party positively. In sum, the political polarization has two types: ideological and emotional. While ideological polarization involves the perception of specific policies in terms of political values, emotional polarization was identified as a sense of group, which indicates hostility toward outsiders. In the real world, it could reflect the desire to avoid the linkage with outsiders.

The literature on the relationship between ideological polarization and emotional polarization presents divergent views. Iyengar et al.'s (2012) study was based on a public opinion poll that found that ideology has a small effect on partisan sentiment, which indicated that partisan membership has a weak ideological background. Dias and Lelkes' research (2021) also confirms that the policy preference didn't drive the inter-party animus. However, Orr and Huber (2019) argued while shared partisanship does influence partisan animosity, the study suggests that policy preferences play a crucial role in shaping evaluations, with individuals showing greater sensitivity to policy positions than party affiliation. Overall, research suggests that political polarization occurs in two distinct forms. The first type is based on political preferences, which are expressed through either support or opposition to policies. The second type relates to identity, influencing personal feelings and attitudes toward other parties.

Polarization didn't appear first on social media. Many years ago, political scientists began to focus on elite polarization. In 1984, Poole and Rosenthal (1984) conducted a study to examine the difference between liberalism and conservatism in the American Senate, and they found a trend where moderate positions disappeared in favor of more extreme positions supported by coalitions. Dalton (1987) investigated how generational changes among political elites contribute to varying levels of ideological polarization. Dalton (1987) found that younger political elites were more ideologically oriented in their political beliefs and had a broader understanding of the policy gap between political parties than older political elites.

The elite polarization prompted the literature to focus on political polarization among the masses. DiMaggio et al. (1996) conducted an analysis of 18 social issues to measure the polarization among the public and found that polarization only occurs in abortion and partisan identities. Subsequently, Evan (2003) used the same method as DiMaggio et al. (1996) and refreshed data and also didn't find evidence of general polarization. Fiona and Abrams (2008) discussed various types of evidence used to study polarization and highlighted that while there is general agreement on elite polarization, public polarization due to irrational empirical strategies is exaggerated. Abramowitz and Saunders (2008) used data from the American National Election Studies and national exit polls and found that ideological polarization has significantly intensified among both the general American public and political elites. The people who labeled democrats become more liberal, and republicans become more conservative. Their argument is that the polarization of political parties in the United States is driven by a small number of activists who can catch public attention (Abramowitz and Saunders 2005). The polarization often depicted in the media cannot represent the general public's attitude since a vocal minority of activists and elites are highly engaged in political discourse and advocacy. These activists actively spread the message to the public that the political landscape on some topics, for instance, abortion, is evolving (Carmines and Woods 2002).

However, all of these findings are based on an era when traditional media still dominated political communication. In that era, political elites were at the center of the communication node and shaped the map of political opinion. In other words, the media helps to amplify elites' views, whether or not they are explicitly political (Prior 2007: 5). Literature further focuses on the relationship between traditional media and political polarization.

As mentioned above, philosophers such as Habermas (2006) argue that the mass media creates unified concerns and thus helps to facilitate integrated public opinion. However, in the empirical realm, researchers identified a trend of media partisanship. Ho and Quinn (2008) collected the editorials about supreme court decisions of 25 major American newspapers from 1994 to 2004 and found that about half of the editorials had a distinct position. For example, the New Yorker Times's editorials tended to be more progressive, while the New York Post bordered on conservative. Groseclose and Milyo (2005) measured media bias by using citations to think tanks and policy groups by both media outlets and members of Congress. They found a liberal bias in some news channels they examined, with only Fox News' Special Report and The Washington Times being

closer to the center. CBS Evening News and The New York Times were identified as having scores far to the left of the average member of Congress, indicating a significant liberal bias. Baum and Groeling (2008) employed a content analysis approach to study the story selection and emphasize the decisions of various news outlets, including wire services, Fox News Channel, Free Republic, and Daily Kos. Their findings revealed differences in story selection among news outlets and shed light on the impact of partisan orientation on news coverage and political discourse in America. Furthermore, by examining COVID-19 vaccine coverage on cable television and mainstream news organizations, Motta and Stecula (2023) found that attitudes toward vaccines varied markedly from one media outlet to another. Fox News' coverage of vaccines became more negative and increased public negativity toward vaccines.

Changes in the information environment can lead to the polarization of political views (Iyengar and Simon 2000). Hopkins and Ladd (2013) analyzed the impact of access to Fox News on vote intentions and political affiliations. Their findings indicated a significant pro-Republican effect among individuals who are more likely to tune in to Fox News, such as Republican identifiers and pure independents. Martin and Yurukoglu (2017) used channel positions as instrumental variables to estimate a model of viewership, voting, and ideology evolution. Their study shows that watching Fox News Channel increases the probability of voting Republican in presidential elections, which indicates that channel positions can influence voting behavior. These findings suggest that media partisanship reinforces the existing political affiliations of its audience. However, when considering a broader demographic, the mass media appears to have mitigated polarization among the general population. Campante and Hojman (2013) found that television and radio reduced political polarization in poorer, less educated areas since they offer a lower barrier to getting information. This can be attributed to the fact that television has brought diverse information and educational impact to entry and increased political awareness in these communities. Jenssen and Aalberg (2019) utilized data from the Norwegian Citizen Panel and found that while Norwegians exhibit selectivity in their news consumption and hold perceptions of media bias, the overall TV news audience in Norway is not highly polarized since Norwegians also consume news from sources perceived as hostile if the content resonates with their political views.

While some literature observed the evidence of partisan media and its impact on election patterns, this is not enough to distinguish between the selection effect and context effect. As Prior (2013) argues, simply observing audience migration does not provide sufficient proof that partisan

media are polarizing Americans. In particular, the observed patterns occur in individuals who might already have partisanship. The more important empirical finding is that most large media outlets are centrist (Groseclose and Milyo 2005; Ho and Quinn 2008). Prior (2013) further argues that the polarization observed in part of the literature is most likely the result of technological advances. Technological advances made it economically feasible to target smaller audience segments. The influence of the traditional media on mass polarization was not found in the wider population (Campante and Hojman 2013; Jenssen and Aalberg 2019). There are two reasons for this: business profits and cross-partisan possibility. Gentzkow and Shapiro (2010) found that readers have significant economic preferences for like-minded news, while firms respond strongly to consumer preferences. In order to reach a wider audience of potential consumers, the mainstream media is less likely to publish extreme views. Moreover, traditional media offer lower access and diverse alternatives (Campante and Hojman 2013; Jenssen and Aalberg 2019). Even the highly partisan viewers do not watch only one channel for a long time.

With the growth of active users, social media has become increasingly the focus of political scientists since the 2010s. Meanwhile, the divergence between Democrats and Republicans has increased rapidly during Obama's presidency (Pew Research Center 2017). Social media has a faster spread and selectable network compared to traditional mass media and is more likely to create political polarization. Although the literature demonstrates the existence of filter bubbles and echo chambers in social media, as stated in the former section, it remains controversial whether social media actually drives the polarization of user interactions (Bruns 2019). Depending on the data and methodology used, two main approaches have been used in contemporary social media polarization research: digital track analysis (data direct from social media platforms) and questionnaire-based experiments (data from a questionnaire).

For social media like Twitter and Facebook, which are highly interactive and networked in nature, researchers always conduct a social network analysis based on follow or retweet data. Barberá et al. (2015) utilized a statistical model that assumed the probability of a connection between two users in a network was negatively related to their distance in a latent ideological space. They estimated the relative positions of users in this latent space using correspondence analysis, which allowed them to identify the ideological positions of social media users. By leveraging observable social networks and users' decisions about whom to follow on platforms like Twitter, they were able to infer individual users' political preferences and estimate their ideological

positions on a large scale. Their study found that Twitter users tend to follow accounts that align with their own ideological preferences, which leads to the formation of ideological echo chambers on the platform. Conover et al. (2011) collected some tweets that include political hashtags and observed that retweet networks on Twitter show high levels of segregation since users more frequently retweet those who share their political affiliations. This indicates that interactions within these networks predominantly reinforce existing beliefs. Bakshy et al. (2015) used the dataset from Facebook, which contains self-reported political stances and shared websites of American users. They measured content alignment by averaging the ideological affiliation of users who shared each article and found substantial polarization in the website addresses that include political content shared by users, with links aligning with the user's political affiliation. Rivero's research (2015) instead focuses on analyzing patterns of following on Twitter and found that users are more likely to follow political accounts that align with their ideological views, substantiating the argument that ideological proximity drives social media interactions, particularly in politically active circles. Similar to Rivers's study, Barberá (2015) also researched the following network on Twitter to estimate the ideology of political elites and ordinary users. By analyzing the connections between users and identifying political actors that users choose to follow, Barberá constructs a Bayesian Spatial Following model to estimate ideological positions along a scale and finds that public exchanges on Twitter tend to occur among users with similar viewpoints, with right-leaning users forming a cluster of highly motivated individuals dominating public conversations. Himmelboim et al. (2013) collected the data based on ten topic keywords, such as tea party and unemployment benefits. Their finding showed that only 42.34% of the 5,000 messages collected were associated with the main clusters, indicating limited exposure to diverse viewpoints. Garimella and Weber (2017) focused on both following and retweeting networks on Twitter. They gathered the followers and retweets of critical political actors on Twitter and, through application program inference, got the duration of following. Their study revealed changes in follow and retweet networks over time and indicated a relative increase of 10%-20% in political polarization on Twitter, particularly during elections.

Some studies further focus on specific topics. Karlsen et al. (2017) focused on online debates related to gender equality, specifically examining how individuals engage with contradicting and confirming arguments in discussions on this topic. The findings of their study indicated that both confirming and contradicting arguments in online debates could lead to a reinforcement of

individuals' original opinions and beliefs. Participants reported emerging from online debates with stronger convictions in their initial viewpoints, regardless of whether they were contradicted or supported by other debaters. Falkenberg et al.(2022) collected tweets around the Conference of the Parties (COP). The analysis revealed a significant increase in ideological polarization during COP events, with a particular focus on COP26. Jiang et al. (2020) utilized a Twitter dataset containing COVID-19-related tweets in the United States. The study found that political partisanship significantly influenced online conversations about COVID-19. Users in Democratic- and Republican-leaning states engaged in discussions that were polarized along partisan lines, with topics often revolving around political issues. Republicans were less likely than Democrats to perceive the pandemic as a serious threat, leading to differences in beliefs and behaviors related to the virus. Treen et al. (2022) used Latent Dirichlet Allocation to identify the topics from climate discourse on Reddit. Their study on climate discourse on Reddit found evidence of polarization in the discussions surrounding climate change since some communities on Reddit were dominated by particular ideological viewpoints.

Not all studies confirm the existence of polarization on social media. Nyhan et al. (2023) conducted an experiment on consenting Facebook users to explore the effect of like-mind information on Facebook. Participants were randomly assigned to treatment and control groups to measure the effects of the intervention. In the treatment group, the contents from the following network were decreased. While the intervention substantially reduced exposure to content from like-minded sources on Facebook, they found there is no measurable effect on political attitudes, affective polarization, ideological extremity, or opinions on issues due to the reduced exposure to like-minded sources. Thus, although most of the content seen by Facebook users comes from the following network, their study suggested these contents have no effect on political polarization. Guess et al. (2023) aim to investigate the effects of reshared content on Facebook during the 2020 US election. In the control group, the users' feeds maintained no changes, and in the treatment group, the reshared contents were removed from users' feeds. Participants in the treatment group saw substantially less reshared content compared to the control group. They did not find any evidence that supports significant shifts in users' political attitudes or behaviors, except for a decrease in news knowledge within the sample. Their finding highlighted that while reshared content on Facebook can increase exposure to both reliable news and content from untrustworthy sources, the removal of reshared posts did not lead to significant changes in political attitudes or

behaviors. Although the experiment can estimate the effect of information exposure on social media on polarization by treatment and control groups, the reliability of the results of the experimental methodology is questionable since researchers cannot have full control over the participants' information sources.

Political polarization on social media could also be related to the platform's nature. Yachi et al. (2021) conducted a cross-platform analysis on Twitter, Facebook, and WhatsApp and found that depending on the platform, the polarization is different. While the polarization increased on Twitter over time, the homogenized interaction on WhatsApp significantly decreased. Yachi et al. (2021) argue that Twitter's interactive nature allows users with very different opinions to interact without needing common goals or identities. This leads to intense debates and increasing division. Similarly, Cinelli et al. (2021) also found significant differences across platforms in terms of homogeneous patterns of network structure and biases in the dissemination of information to like-minded users. Ribeiro et al. (2020) measure polarization by analyzing the commenting behavior of users across different ideological communities on YouTube channels. Their finding reveals a trend towards polarization, where users are progressively engaging with more extreme content and ideologies. This polarization is evident in the overlapping user base across different communities and the migration of users towards increasingly radical content over time.

Social media platforms such as Facebook and Twitter exacerbate political polarization, with interactions occurring more often between like-minded individuals (Barberá et al., 2015; Conover et al. 2011; Bakshy et al. 2015). The interactive and selective nature of social media tends to amplify communication isolation. However, research offers conflicting findings on whether social media directly increases political polarization. Some studies suggest it has no significant impact on users' political attitudes or behaviors (Barberá 2014; Guess et al. 2023; Nyhan et al. 2023). The nature of the social media platform also plays a role in polarization (Yachi et al. 2021). In some cases, exposure to social media can intensify polarization, while in others, it may not have a noticeable impact or could even reduce polarization. Most research has focused on platforms like Twitter, where data is more accessible, but other platforms like YouTube, which also play a critical role, have been largely overlooked. Due to YouTube's vast user base and the prevalence of video content, more research is needed to understand its impact on political polarization.

While exploring political polarization, it becomes evident that similar mechanisms also influence discussions and perceptions of economic inequality. Just as social media can intensify or mitigate

political divisions, it also shapes how issues of inequality are highlighted or obscured. This transition is particularly crucial as economic inequality has become a growing concern in contemporary democracies.

2.4. The narratives on income inequality

Inequality has been a core political theme for a long time. Bobbio (1996) argues that the political divergence represents “attitudes of individuals within the societal construct towards the paradigm of equality.” In recent years, income inequality in developed countries has received increasing attention. According to Piketty’s research (2014), economic inequality in Western countries has been increasing rapidly since the Industrial Revolution due to capital accumulation at a much faster rate than economic growth. Saez and Zucman’s research (2020) based on distributional national accounts shows significant disparities in income and wealth distribution in the United States. The top 1% of earners have seen their income share rise substantially since the 1970s, while the bottom 50% have experienced stagnant income growth. Wealth inequality is even more pronounced, with the top 1% holding a disproportionately large share of total household wealth. Income and wealth inequality are closely intertwined, with changes in income distribution impacting wealth distribution and vice versa. According to Kuhn et al. (2020), an important reason for the rise in income inequality is the unproportional increase in investment returns and wages. In easy words, capital gains make the rich richer and richer, while the wage-dependent working class is suffering from the effects of inflation. The rise of research on inequality has been accompanied by a growing interest in developing narratives that explain why inequality is reasonable or becomes more extreme. As Atkinson (1983) points out, “Different principles of justice lead to quite different views about inequality.” The view of inequality is influenced by differences in perceptions. However, there is almost no literature that summarizes these different narratives.

Some scholars view the persistence of inequality as an inherent irrationality within capitalist democracies. The first narrative about economic inequality is the structural problem of capitalism democracy (Reich 2016). The argument goes that inequality is a consequence of the increasing concentration of political power in the hands of corporate and financial elites who have more influence on economic policy. This argument also argues that markets are artificially shaped and that inequality results from market rules that favor elites. Case and Deacon (2021) argue that a significant portion of the American economy is controlled by the wealthy, often with the support of

the government. They attribute much of the wealth at the top to unfair or illegitimate means. The focus is not solely on inequality but on the system's unfairness. The second narrative believes that social inequality comes from the failure of the capitalist market (Stiglitz 2012). Stiglitz (2012) also argues the rent-seeking of elites is responsible for social inequality. However, he observed an important phenomenon in the American economy: instead of driving up the wages of the working class, increased productivity made their situation worse. According to economic principles, wages are determined by the relationship between supply and demand. Higher productivity drives down aggregate demand in society, and lower aggregate demand leads to fewer jobs in certain industries. These retrenched workers are forced to seek their livelihoods in other industries. As supply in the labor market increases, wages fall. Stiglitz contends that such market failures necessitate government intervention and policies aimed at redistribution. The third narrative for inequality emphasizes the disconnect between wage growth and profit growth, especially the long-term stagnation of wages for workers at the bottom, while incomes and incentives for higher-income groups have risen sharply. Mishel and Davis (2015) found that chief executive officers from top companies earn 300 times more than regular workers. Piketty (2014) questions whether such a gap in income really represents the difference in efficiency. In other words, do executives really generate hundreds of times more revenue for their companies than an average worker? The fourth narrative (Piketty 2014) blames injustices in the process of capital accumulation, including such issues as the passing of inheritance and the fact that tax rates on capital income are usually lower than those on labor income. The fifth narrative attributes inequality to changes in external contextual factors, such as globalization and technological advances.

Economic narratives that justify inequality often depict it as an inevitable outcome of economic development and market competition. Kuznets (1963) discovered a curve that looks like an upside-down U curve when he plotted income inequality against GNP per person. According to Kuznets' finding, income inequality is likely to rise during the early phases of a country's development when people are moving to cities and industries are growing. The most famous view justifying income inequality comes from neoliberalism economists Hayek and Friedman, who believe that the market makes the best decision. Hayek (1969) proposed a market-centric theory of price determination, arguing that market forces, through supply and demand mechanisms, establish prices more efficiently than government intervention. He further contended that an individual's economic value is directly reflected by their market productivity, as determined by their willingness and ability to

contribute within a free market system. Hayek further contended that prioritizing absolute equality through centralized planning and regulations fosters an oppressive social structure, potentially culminating in dictatorial rule and individual subjugation. Finally, Hayek advocated for minimal government regulation, positing that unfettered free enterprise fosters optimal economic outcomes. Friedman (1963) posits that the wealth distribution generated by market forces and capitalist exchanges is both equitable and beneficial. He acknowledges that these mechanisms lead to inequality but argues that this is a necessary and acceptable byproduct of the economic freedoms inherent to capitalism. In market competition, each individual is free to act as he or she wishes and should, therefore, be responsible for adverse consequences. Furthermore, Friedman contends that attempts to rectify this perceived inequality through interventionist policies may ultimately lead to greater injustices. Another narrative thinks inequality is the result of differences in individual efficiency. Sowell (2016) criticizes discussions on distributional inequality for overlooking the efficiency of individual productivity. He argued that more productive individuals enhance overall economic productivity by contributing valuable human capital, which benefits society as a whole. Sowell emphasizes that equal opportunity should not be mistaken for equal outcomes and that the accomplishments of certain groups should not be viewed as privileges. In his mind, the wealthy's accomplishments are seen as the result of their talent and efforts. Similar to Sowell's argument, Herrnstein and Murray (1994) believe inherited differences in intelligence explain inequality. They argue that intelligence determines many aspects of life, such as educational attainment, income, job performance, and social status. One additional justification is inequality promotes economic growth, which is represented by trickle-down economics (Again and Bolton 1997). The narrative of trickle-down economics posits that benefits given to the wealthy or advantages enjoyed by the top earners can stimulate economic growth that will eventually trickle down to the lower-income segments of society. The critical point is not inequality but poverty avoidance. This theory supports policies that favor tax breaks, deregulation, and financial benefits for corporations and the wealthiest individuals under the assumption that their increased spending and investment will benefit the broader economy.

In recent years, left-wing politicians like Bernie Sanders and Alexandria Ocasio-Cortez have always used Twitter to wake up the public to the current state of inequality in America. The growing engagement with economic inequality on social media reflects a significant intersection between academic theories and public discourse. Hashtags such as #WealthGap, #EconomicJustice, and

#TaxTheRich illustrate how users rally around the concepts discussed by scholars like Reich (2016) and Stiglitz (2012). Although there are many debates around income inequality in the real world, the academic literature has largely remained silent on how these debates unfold on social media platforms. Considering the rising role of social media in opinion formation, it is important to study how discussions about economic inequality are happening on these platforms.

2.5. Conclusion

The literature reviewed discusses social media's impact on modern democratic processes and public discourse, particularly the complex phenomena of selective exposure and political polarization. In addition, different narratives of inequality are presented.

First of all, social media has shifted the public sphere from an elite-dominant, passive-consumption model to an interactive network. Unlike traditional media, where content is crafted by professionals and audiences receive it passively, social media platforms empower individuals to actively participate in content creation and dissemination s (Benkler 2006; Thompson 2005). This democratization of media communication has allowed individuals to engage in public conversations every day (Bruns and Highfield 2015). However, it has also led to the fragmentation of the public sphere, as people increasingly focus on niche interests rather than a collective public agenda. This fragmentation raises significant concerns (Habermas 2023) about the ability of social media to foster unified discourse and effective consensus building, which are critical for the health and efficacy of democratic deliberation.

Social media also empowers users to choose more information. This self-confirming exposure to information may result in ideological segregation on the Internet, i.e., people will only see information that is politically consistent. For example, An et al. (2013) found that Facebook users tend to consume news that aligns with their political stance. Mukerjee and Yang (2021)'s finding suggests that users on social media avoid information they disagree with. While research confirms the existence of information segregation on social media, however, the scale of this segregation has been questioned. Some scholars (Fetcher et al. 2021; Nelson et al. 2017) argue that selective information exposure exists only in a small proportion of the population.

The consequence of selective information exposure is that social media users interact almost exclusively with others who share their ideology. This led to the formation of emotional polarization on social media. In the case of emotional polarization (Baldassarri and Page 2021), individuals'

attitudes are highly correlated with partisan labels and even spread to other non-political domains, with high identification with in-group members and hostility towards out-group members. Empirical evidence shows that social media users tend to interact within groups that are politically aligned, both in terms of following networks (Barberá et al. 2015; Rivero 2017; Garimella and Weber 2017) and retweeting or resharing patterns (Conover et al. 2011; Barberá 2015; Bakshy et al. 2015). Although social media tracking data show evidence of polarization of group interactions, experimental methods question whether polarization exists in the real world. By removing reshared content or content from the following network, the researchers (Nyhan et al. 2023; Guess et al. 2023) compared differences in political attitudes between the treatment and control groups. They found no significant change in the political attitudes of Facebook users after intervening in the content they saw. While these findings challenge the role of social media in ideological polarization, the validity of the results is questionable since the experimental method could not isolate social media as the sole source of news.

The narratives on income inequality reveal diverse perspectives on its causes and implications. Scholars such as Reich (2016) and Stiglitz (2012) argue that economic inequality stems from structural problems within capitalist democracies and market failures, while others like Hayek (1969) and Friedman (1963) view it as a natural and beneficial outcome of market competition. Despite varying views on the reasons behind inequality, there is consensus on its persistence and the need for deeper analysis. However, the academic literature has largely overlooked how these debates unfold on social media, highlighting the necessity for research on this evolving dynamic. Considering the influential role of social media in shaping public opinion, it is crucial to examine how discussions on economic inequality are conducted on these platforms.

The extent to which existing empirical research addresses theorists' specific concerns remains questionable. Habermas (2006) worries that social media fragmentation prevents the public sphere from providing stable political input to democracy. Regardless of the methodology, current studies often categorize social media content into two political extremes: left and right. This simple dichotomy overlooks the depth of content analysis. How do the words used by different groups on the same topic differ? How does this disparate use of narratives impact common understanding? For example, while narratives about income inequality can be divided into left-wing and right-wing based on attitudes toward inequality, these narratives differ considerably in their understanding of inequality. Additionally, there are variations in language intensity within the same political position.

For instance, on the topic of inequality, some discourses strongly support redistribution, while others are milder. How do these differences influence social media users' consumption patterns?

Much of this research gap is because existing research is overly focused on platforms such as Facebook and Twitter. On these platforms, retweets and follows constitute the main form of media consumption. However, whether the observed patterns apply equally well to other platforms with a different media consumption model where research is underrepresented remains unknown. YouTube is the world's largest and most influential streaming platform. Whether an interactive polarization similar to that found in social media such as Twitter and Facebook can be found in more content-rich video platforms remains a question worth investigating.

Considering the growing economic inequality, this study aims to fill the research gap and provide insights into how social media shapes discussions on this topic. In the following, this study focuses on the YouTube platform and provides an in-depth analysis of the discussion of income inequality. By combining video content and comments data, this study explores how narrative differences in videos, the degree of polarization in interactions, and the intensity of viewpoints affect viewers' perceptions and interactions.

3. Theoretical framework

In recent years, YouTube has become a major platform for video distribution and interaction, fundamentally changing how people consume and discuss information online. This chapter explores the unique features of YouTube in three dimensions: content production, distribution, and consumption. Based on these characteristics, the chapter introduces three research questions in detail and corresponding hypotheses to investigate the dynamics of YouTube's interaction networks and their implications for public discourse.

3.1. YouTube and its nature

YouTube was founded in June 2005. Its original purpose was to eliminate technical obstacles that hindered the broad dissemination of videos on the internet (Burgess and Green 2018). In April 2024, YouTube reached 2.5 million monthly active users, behind only Facebook. Now, YouTube is the most important and largest online video platform.

Like most social media platforms, YouTube allows users to engage with content through likes, comments, and shares. This interaction mechanism helps create a community feel and facilitates user discussions. YouTube users can create personal profiles, much like they can on platforms like Facebook or Instagram. These profiles can be customized and used to upload videos, create playlists, and more. Additionally, users can subscribe to channels they like, similar to following networks on Twitter and Facebook. In addition, YouTube empowers users to be content creators, not just consumers. Users can upload their videos and create content series. This aspect resembles platforms like Instagram and TikTok, where user-generated content is the core of engagement.

Compared with other social media platforms, YouTube has its unique features. Regarding production dimension, video format contains more content than text format. It allows for in-depth exploration of complex topics, such as podcasts or debates, which provides a robust foundation for content analysis. Gruzd and Roy (2014) suggest that while Twitter exposes users to diverse viewpoints, the platform may not always facilitate meaningful discussions due to the constraints of character limits. Regarding the distribution dimension, YouTube does not rely on follow-and-share mechanisms to distribute content, and it does not have a traditional social interface like Facebook and Twitter. YouTube distributes content primarily through recommendation algorithms and video searches rather than relying on direct social interaction between users. YouTube differs significantly

from other social platforms in terms of media consumption. While other platforms (e.g., Twitter and Facebook) consume content primarily through likes and retweets, YouTube's user interactions focus more on video comments. On platforms such as Twitter, the heat of content is updated in real-time, and videos can be viewed as long-term information nodes on YouTube due to their relatively low interactive properties. The comments under each video constitute a separate sub-network with more persistent discussion and interaction properties.

3.2. Hypothesis development

Klinger and Svensson (2015) argue that the core of production logic on social media is to attract more attention. To achieve this, content should be highly personalized to target specific groups of people. Meanwhile, social media empowers everyone to produce and distribute content, leading to a proliferation of voices. Dahlgren (2015) contends that this democratization of content creation has fragmented the public sphere, as different groups focus on different issues. According to Habermas (2023), this fragmentation undermines democracy by failing to direct citizens' attention to critical issues and preventing the emergence of a quality consensus necessary for informed decision-making.

Previous studies have tended to measure the fragmentation of discussions on social media primarily in terms of political positions, categorizing content into two segments—left and right. This approach neglects to analyze the richness of the content, especially the underlying narratives that shape and influence these discussions. For instance, there may also be different narratives on income inequality within the same political position.

Research Question 1: How divided is the discussion of income inequality on YouTube?

Habermas (2006) argued that the structuring principles of public discourse and the process of information transfer have been radically altered because of the expanded amplitude of digital media. This digital revolution has brought multiplication in media sources and further democratized production. While increasing access, this has led to the fragmentation of speech. The progressive elimination of the traditional gatekeeper role has given way, as Bruns and Highfield (2015) describe, to “produsage”: a phenomenon in which content producers and consumers blur, hence allowing disparate and sometimes even oppositional narratives to coexist within the same space of the digital. Therefore, the concept of the issue public - a notion of solitary groups acting politically in reaction to specific issues or worries - takes on here a new meaning. Dahlgren (2015) points out

that with the emergence of the internet age, completed public space has been removed in order to give into the tendencies of groups. YouTube, the biggest streaming platform in the world, has a large user base. These groups will be free to generate and share content on YouTube, venting their views, which might lead to narrative disintegration.

Hypothesis 1: Discussions on YouTube on social inequality exhibit clear narrative division in terms of content.

This hypothesis is based on the observation that YouTube lacks traditional gatekeeping, and its decentralized content creation system naturally allows for a broad spectrum of narratives. These narratives could stem from the creators' beliefs, geographic and cultural backgrounds, political ideologies, and target audiences. Such factors may guide creators in developing video content that aligns with their political views or those of a particular group.

According to Baldassarri and Page's definition (2021), political polarization encompasses ideological and emotional polarization. While ideological polarization pertains to differing political beliefs, emotional polarization is rooted in group identity and loyalty. Differing views on politics lead to the forming of distinct group boundaries, fostering a strong sense of in-group and out-group dynamics. This polarization results in varied attitudes towards members of one's group compared to those of the opposing group, with interactions more likely to occur within politically like-minded groups. For example, on Facebook and Twitter, following occurs between users who share the same position. However, due to different media consumption patterns, whether the polarization observed on Facebook and Twitter is also present on YouTube's interaction network remains unclear.

Research Question 2: How polarized are the interaction networks of videos about income inequality?

Social media allows users to choose who they communicate with, enabling them to surround themselves with like-minded voices. Historically, even with mass media, people preferred newspapers that aligned with their political beliefs (Gentzkow and Shapiro 2010). Thus, it is possible that the same homogenized interaction patterns observed on Twitter and Facebook are also present in YouTube's interaction network.

Hypothesis 2: The comments show a clear homogenization, with users tending to interact with people who share their political stance.

Hypothesis 2 would have it that comment sections on YouTube present an opinion-based form of homogeneity, where persons are more likely to engage people holding similar political and

ideological standpoints to their own, particularly on some topics, as seen with dimensions of social inequality.

An echo chamber is an enclosed media space that amplifies and insulates messages, creating reinforcing feedback loops and a common frame of reference for its audience without opposing perspectives (Jamieson and Capella 2008). According to existing definitions (Sunstein 2007; Jamieson and Capella 2008; Nguyen 2018), echo chambers have two main characteristics: closure and a network of similar views around a central point. Previous studies on Twitter and Facebook often conflate polarization with echo chambers, as the following networks constitute the primary mode of content consumption on these platforms. On YouTube, videos serve as persistent information nodes, while comment sections provide a relatively closed environment for interaction. Given this context, it is important to investigate how video content on YouTube contributes to the formation of echo chambers.

Research Question 3: To what extent does video content attract comments of the same political stance?

YouTube's recommendation algorithm, content creators, and user engagement dynamics all significantly shape how information is disseminated and received on the platform. A video's subject matter and tone may attract viewers with specific interests or beliefs. For example, a video that takes a definitive stance on a controversial issue like social inequality might predominantly attract viewers who agree with that stance, unlike a more neutral or balanced presentation, which might invite a broader spectrum of opinions.

Hypothesis 3: The homogeneity of comments on YouTube videos is strongly influenced by the explicitness of the content's ideological orientation.

This hypothesis responds to the echo chamber argument. YouTube videos with strongly expressed ideological stances, particularly on controversial issues like social inequality, tend to attract more homogeneous comments - meaning that the comments generally align closely with the perspective presented in the video.

Two points from the literature support this hypothesis. Firstly, users tend to select content that confirms their pre-existing beliefs, a phenomenon known as selective exposure. Mukerjee and Yang's study (2021) shows that no matter how popular news is on social media, users typically avoid news from sources they disagree with politically and prefer news from sources they agree with. Similarly, Cinelli et al. (2020) found users only interact with a very small number of online

news organizations. The second factor is the recommendation algorithm. In order to attract more active users, all major social media platforms adopted personalized recommendation algorithms where users are increasingly exposed to information that aligns with their existing beliefs and preferences. (Pariser 2011).

4. Data

In this research, a custom Python application interfaced with the YouTube Data Application Program Interface (API) to collect data related to the keyword of income inequality as part of the investigation into the representation and discourse around socioeconomic issues on digital platforms. Python is a programming language that is widely used for data collection and analysis due to its powerful libraries. With a unique API key, the Python application can access YouTube's endpoints to query data, including the search list, video list, comments list, and replies list for individual videos.

4.1. Data collection

Data were systematically collected using different uniform resource locators (URLs) from API to ensure a streamlined access process. The acquired data were organized into a data frame, which supported structured storage and simplified further analysis.

The data collection process was divided into two steps: collecting videos and their comments. In the first step, a loop was implemented to query the YouTube API for videos related to the keyword "income inequality." The results returned by the API are consistent with any user's search on YouTube. Each API request fetched 50 video results (the maximum limit set by the API) on one page, and pagination is handled through pageToken, which helps navigate search results across different pages. The loop continues until no next page token is returned, indicating no further pages are available. For each video, the titles, the video IDs, the description, the comment count, the view count, and the publish time were extracted. Meanwhile, channel statistics are obtained, which provide data such as the number of subscribers.

The video's preferred content region is restricted to "US", which means that users in the United States can see these videos. There are two reasons for doing so. Firstly, the scope of the study is limited to a single society. The United States has, in recent years, been one of the most unequal and politically polarized countries in the world. Secondly, as the birthplace of YouTube, there exists a huge number of users in the United States who can offer enough content for research. The videos collected contain content about not only the United States but also India, the Netherlands, the United Kingdom, and Singapore, among other places. Considering the immigrant background of American society, these videos were retained in the study.

The inclusion criteria for the dataset specify that only videos with more than ten comments are selected. This decision is based on two key considerations: firstly, videos with few comments did not receive significant public interaction, which is crucial for this study's focus on social discourse. Secondly, a low comment count often correlates with lower video quality to the audience, which could skew the content analysis. Therefore, filtering out videos with minimal comments ensures the data collected is both relevant and sufficient for analytical purposes. Furthermore, repeated videos in the dataset were removed. After filtering videos, the dataset contains 276 videos.

Subtitles were subsequently collected using the yt-nlp package, a downloader specifically designed for retrieving YouTube subtitles based on the video IDs. Through the yt-nlp package, subtitles for 225 videos were successfully downloaded, while the subtitles for 51 videos could not be downloaded successfully. This failure was attributed either to a mismatch in the subtitle format or to the absence of subtitles on the videos. After gathering the subtitles, timestamps were removed with regular expressions.

Moreover, the quality of some subtitles was found to be inadequate, as they were provided by the creators themselves. The YouTube audio was transcribed using Maestra¹, which employs artificial intelligence to transform video into text. All the re-generated subtitles in the dataset were marked accordingly.

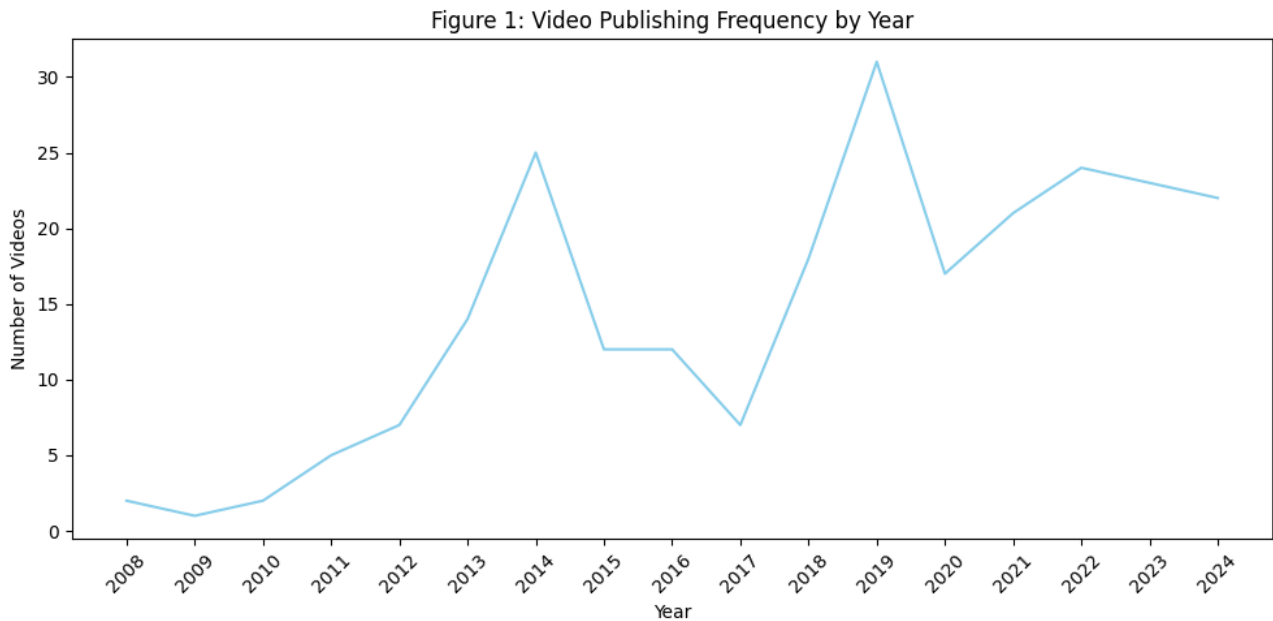
After the collection of subtitles was completed, I checked the videos' subtitles to ensure that the content was in line with the direction of this study. Thirty-one videos are not related to the research's purpose. These videos were labeled and filtered after the data was processed. All data in the first step collected was saved as a CSV file with an index via the pandas package, and the data collection was completed on April 9th.

The second step is to collect all the comments below the videos. Since the YouTube data API v3 allows a 110,000 quota daily, the comments were collected over four days from April 9th to April 11th and April 22. The video ID was saved in the first step's CSV file, and the collection was completed through two endpoints of YouTube API. Firstly, the endpoint CommentThreads was used to collect the primary comments that directly reply to the video. Secondly, the endpoint Comment was used to collect secondary replies that responded to primary comments. The unique comment IDs for primary comments were saved in the dataset, and secondary replies were marked as true in column is_reply. Similar to the first step, page turns were set until there was no next page. Python

¹ More information available at: <https://app.maestra.ai/>

application thus traversed and saved all comments and replies. The results of the comments collection were saved as 276 files by video index after filtering for duplicates. Finally, 341,556 comments from 245 videos were collected after removing irrelevant videos in the first step.

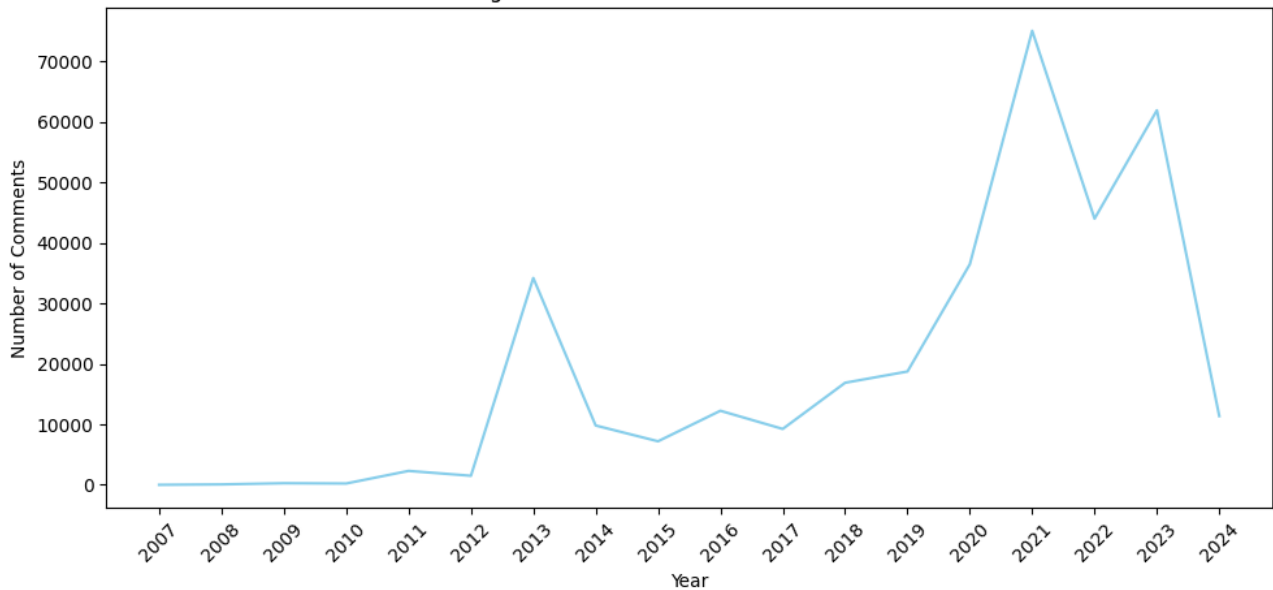
4.2. Data description



Figures 1 and 2 illustrate the trends in public engagement with the topic of income inequality on YouTube up to April 2024. Figure 1 displays the annual distribution of video releases concerning income inequality, while Figure 2 presents the yearly volume of comments on these videos. The line charts depict two distinct waves in activity. Figure 1 identifies 2014 and 2019 as the years with the highest number of video releases related to income inequality. According to the World Bank data, these peaks coincide with the U.S. Gini index reaching its apex at 41.5. Figure 2 demonstrates the frequency of interactions with videos on income inequality. The peak periods for video interactions occur in 2013 and 2021. This indicates a shift in viewer engagement compared to the years of highest video production.

The observed shift in peak years for video releases and interactions can be attributed to two reasons. Firstly, inequality in the United States became a popular topic when Piketty published his book “Capital in the Twenty-First Century.” The discourse around Piketty’s book captured substantial audience attention and fostered significant public discourse. This heightened public engagement influenced content creators and led them to amplify their focus on income inequality.

Figure 2: Number of Comments Per Year



Consequently, this resulted in an increased production of related videos during 2014. This phenomenon illustrates a responsive dynamic between public issues and content creation, where societal events catalyze content producers to address emerging or escalating topics. The peak of interaction in 2021 was associated with block-down and home offices due to the COVID-19 pandemic. As people spent more time online, the availability of time combined with the heightened concerns about economic disparities - exacerbated by the pandemic's impact on employment and social mobility - led to increased viewing and interaction with videos discussing income inequality.

Figure 3: Number of Comments per Video (Capped at 10,000)

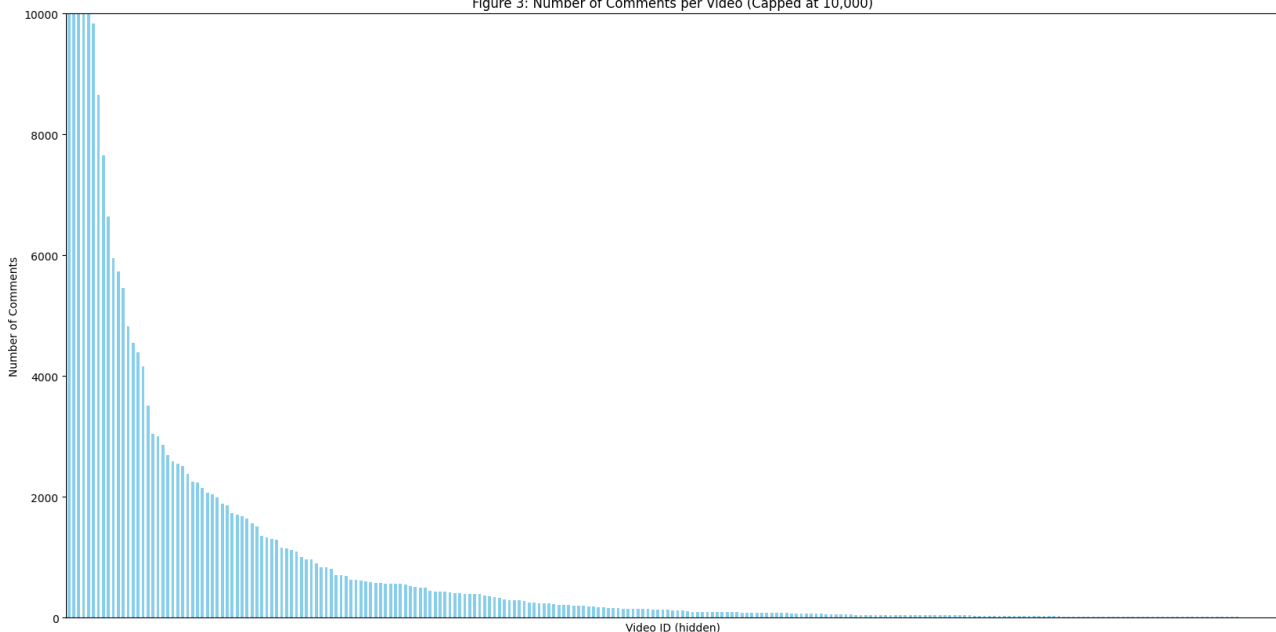


Figure 3 shows the distribution of video comments. Owing to the wide disparity in counts, the Y-axis is truncated at 10,000 to better show quantitative distribution. There are five videos with comments above 10,000, and the largest of these videos reached 65,000. The presence of a long-tailed distribution can be seen in the figure. Although there is a steep peak to the left, most videos contain a very low amount of comments and form a long tail. This pattern suggests that while a few videos attract significant audience interaction, most videos receive relatively minimal.

Figure 4: Proportion of Replies in Comments

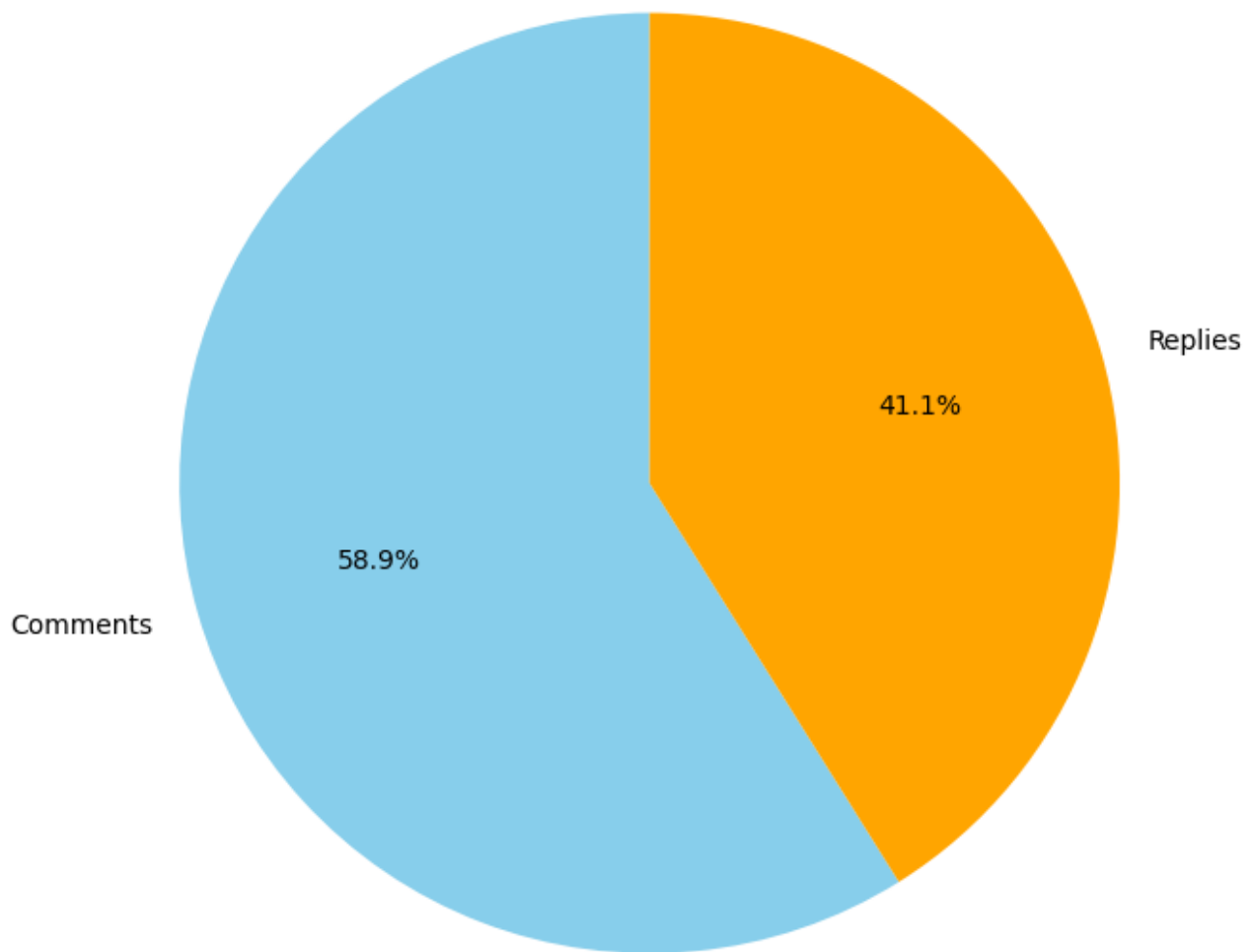
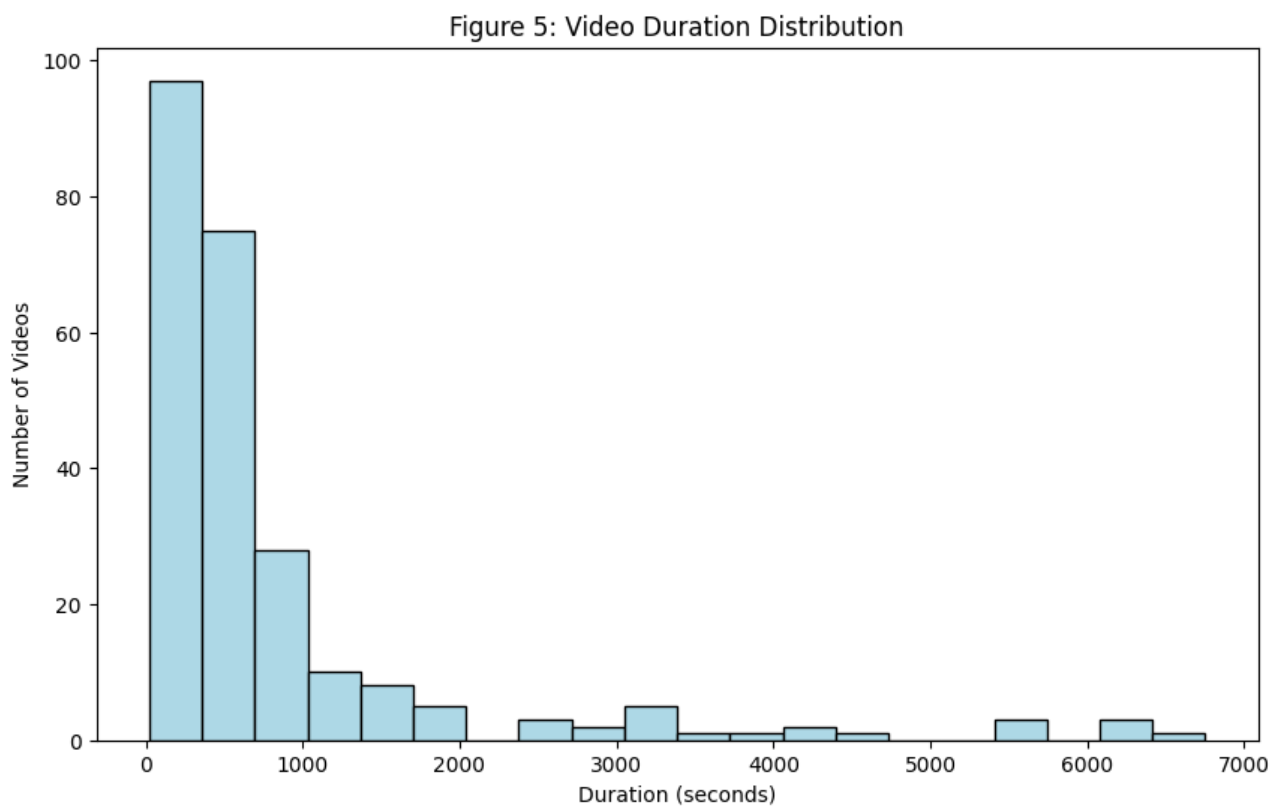


Figure 4 shows the dataset's relative distribution of primary comments and secondary replies. The larger segment, shaded in blue, represents original comments, accounting for 58.9% of the total. The smaller orange segment represents replies to those comments, constituting 41.1% of the total.

dataset. This chart suggests a robust level of interaction engagement on YouTube, where almost as many replies are made as original comments.

Figure 5 illustrates the distribution of video durations in seconds, showing a clear right-skewed pattern. Most videos are less than 1,000 seconds (about 16 minutes) long. Beyond 2,000 seconds, the number of videos stabilizes at lower values, indicating that while longer videos are less common, they still exist. This distribution reflects the diversity of video content. Longer videos may include podcasts and documentaries, while shorter videos are likely news reports. YouTube's content contains significantly more information than Twitter, which is limited to 140 characters.



5. Analysis of content production: The narrative divergence

Previous research indicates that social media platforms, particularly YouTube, enable a decentralized and democratized content creation environment, which can lead to a fragmentation of discourse. According to Klinger and Svensson (2015), the logic of content production on social media is centered around attracting attention through highly personalized content targeted at specific groups. The concept of “produsage,” described by Bruns and Highfield (2015: 57), encapsulates the blurring of lines between content producers and consumers, allowing for diverse and sometimes conflicting narratives to coexist within the digital space. This chapter aims to answer the following question: How divided is the discussion on the topic of income inequality on YouTube? Based on the existing literature, it is hypothesized that discussions on YouTube about social inequality exhibit clear narrative division in terms of content.

To explore this narrative divergence, this study employed the BERTopic package (Grootendorst 2022), a document embedding method based on a pre-trained transformer language model. The methodology section details this method, explaining the reasons for its selection and its limitations. The results section then discusses the output and explores possibilities for future research.

5.1. Methodology

Although narratives have been recognized as having a significant influence on human behavior and decisions (Schiller 2017), the methods for systematically analyzing and extracting insights from these narratives remain underdeveloped. Ash et al. (2024) introduced a method that integrates three traditional unsupervised techniques: semantic role labeling, named entity recognition, and word embedding. Despite its novel framework, this approach has shortcomings in extracting narratives effectively. The method begins with semantic role labeling, simplifying sentences by retaining only their main constituents. However, the strength of transformer language models lies in their ability to understand context deeply. This simplified sentence structure hinders the model’s ability to distinguish between proxemics and antonyms (Vahtola and Tiedemann 2022), leading to numerous

counterfactuals and meaningless narratives². Consequently, while the approach shows promise, it struggles to distinguish narrative differences within highly similar texts.

Another way to obtain narratives is topic modeling, where narratives in a text are obtained by interpreting topic words. The core idea of this method is that if there is a difference in the narratives of income inequality across videos, then this difference should be reflected in the words they use. There are three traditional topic modeling methods: Latent Dirichlet Allocation (Blei et al. 2003), Latent Semantic Analysis (Valdez and Goodson 2018), and Structural Topic Modeling. These traditional topic modeling techniques offer a distinct approach to narrative extraction by assuming that documents can be decomposed into a mix of underlying topics, each represented by a collection of words. The application of these methods allows researchers to infer the predominant narratives within a document or corpus by analyzing the distribution and relationship of topics.

However, each of these methods has its limitations. Latent Dirichlet Allocation (LDA) often requires a predefined number of topics, which may not accurately reflect the complexity or the dynamic nature of the narratives within the data. Latent Semantic Analysis (LSA) struggles with handling synonymy and polysemy, which can lead to less coherent topic interpretations. Structural Topic Modeling (STM), while allowing for the incorporation of metadata to improve topic quality, can be computationally intensive and requires robust parameter tuning to achieve optimal results.

The collected video content data has three features. Since the videos are extracted based on the keyword income inequality, it is to be expected that the content of the videos will be highly relevant to inequality. Furthermore, videos on social media are interactive and contain oral expressions. For example, in a video, there are about 30% greetings and introductions. Therefore, compared with written text, information density is not uniform. The core of the video may be in a handful of minutes. The third challenge is related to text processing. The research question requires capturing the narrative divergence in video content, not the topics of all video content. More precisely, we need to find the content differences between videos. Thus, each video must be analyzed as an individual unit of study rather than aggregating them into a single corpus.

This poses a challenge to the method. Traditional methods based on word bag representation and topic prevalence, such as Latent Dirichlet Allocation (LDA), Latent Semantic Analysis (LSA), and Structural Topic Modeling (STM), may not be adequately equipped to address the specific needs of

² For example, in the dimensionality reduction process, the model may oversimplify by merging opposite entities, such as rich and poor, into a single entity.

analyzing video content (Churchill and Singh 2022). These traditional methods are designed to extract common themes from large volumes of text and rely heavily on the frequency and co-occurrence of words, which may not capture the nuances of spoken language and the contextual environment of video content. Another problem with Latent Dirichlet Allocation (LDA) is the need to specify the number of topics in advance. For unfamiliar data, it is often necessary to manually try multiple parameters to get the best solution (Churchill and Singh 2022).

Another advanced way to study the research question is topic modeling with word embedding (Mikolov et al. 2013), such as LDA2vec (Moody 2016) or Topic2vec (Angelov 2020). Word embedding is a technology that can calculate the word's similarity in a document through vectorization. This capability is particularly beneficial for analyzing documents that contain ambiguous or highly specialized language. However, these methods face two challenges. Video text presents challenges related to the sparsity of data. The spoken content can be less structured and more fragmented than written text. This sparsity can lead to less reliable word co-occurrence and semantic similarity metrics, which are critical for the performance of embedding-based models. Secondly, since the videos are highly related to the topic of social inequality, the repeated use of some words or the reiteration of related contexts could cause semantic saturation in parts of the word embedding space. The model will not distinguish well between the concepts because word vectors in these parts are very close. Thus, a new method is required. On the one hand, this method must ignore the irrelevant part of video content; on the other hand, this method should be able to identify sub-topics under the topic of social inequality accurately.

With advancements in natural language processing technologies, new methods now leverage pre-trained language models to enhance the coherence and accuracy of topic modeling. The BERTopic method uses a transformer language model³ to convert paragraphs to vector representation (Grootendorst 2022). BERTopic generates topic representations through a three-step process. Firstly, the pre-trained transformer model computes sentence embeddings for each document. Subsequently, to enhance the clustering process, the dimensionality of these embeddings is reduced. Finally, topics are derived from the document clusters using a tailored class-based variation of TF-IDF to extract representative topic representations. This approach can take advantage of the strengths of the transformer model. The transformer language models were already trained on large-

³ It is important to note that although BERTopic includes BERT in its name, it is not related to the basic BERT model ((Bidirectional Encoder Representations from Transformers) used in the next chapter. It uses the Sentence Transformer model based on the attention mechanism.

size corpora from different sources, which allowed them to capture the subtle semantic relations of words and sentences even when the texts have high semantic similarity. This enables BERTopic to understand the content and context of documents more accurately and generate more accurate embeddings and clusters.

While BERTopic can use a pre-trained language model to enhance contextual understanding, it cannot automatically ignore irrelevant content. This can result in irrelevant parts being clustered separately from the main topic during modeling. Consequently, only two topics might emerge due to significant differences: one focused on inequality, the selected theme, and another on textual noise. To better achieve the research goals, a caliper-like approach was used to filter out sentences not relevant to the topic.

Two word sets were provided for this purpose: one related to politics (containing words like politics, politician, institution, and ideology) and another related to inequality (containing words like inequality, wealth, income, and market). These word sets were designed to cover various situations related to narratives of income inequality. For instance, discussions might include the impact of ideological policies implemented by specific politicians, such as Reagan and Thatcher's neoliberal policies. Additionally, there may be direct discussions on how markets impact wealth and income inequality.

I used the sentence transformer model, all-mpnet-base-v2 (Reimers and Gurevych 2019), to convert the word sets and each sentence in the video content into embedding vectors. The all-mpnet-base-v2 model, trained on more than 1 billion training pairs, is currently the best-performing universal model for sentence embedding tasks⁴. Next, I calculated the cosine similarities between the word sets (politics and inequality) and each sentence. Cosine similarity measures the degree of similarity between two vectors, ranging from -1 (completely disjoint) to 1 (completely correlated). A threshold of 0.2 was set, which is a low barrier, to filter out sentences unrelated to the word sets. If a sentence had a cosine similarity higher than 0.2 with either word set, it was retained. This approach minimizes the influence of personal judgment on the research outcomes by applying a standardized filtering method, enhancing the objectivity and reliability of the findings.

Table 1 below shows a sample of filtering.

⁴ According to Sentence Transformers, all-mpnet-base-v2 provides the best quality. See: https://www.sbert.net/docs/sentence_transformer/pretrained_models.html

Filtered video text	Removed sentences
In South Africa, one of the most unequal countries in the world, the richest one-tenth of 1%, owns almost 30% of all the country's wealth, more than double what the bottom 90% owns. Income and wealth inequality are not new. In fact, economists and historians who've charted economic inequality throughout history haven't found a single society without it. Which raises a bleak question: is inequality inevitable? One way to estimate inequality is with a number called the Gini index, which is calculated by comparing the income or wealth distribution of a perfectly equal society to the actual income or wealth distribution. A Gini of 1 indicates perfect inequality - one person has everything and everyone else has nothing. A Gini index of 0 indicates perfect equality - everyone has exactly the same income or wealth. Economic inequality is deeply entangled with other types of inequality: for example, generations of discrimination, imperialism, and colonialism created deeply rooted power and class inequalities that persist to this day. But we still need at least a rough measure of who gets how much in a country. Some countries are, economically, much more unequal than others. And that's because a significant portion of economic inequality is the result of choices that governments make. Let's talk about some of these choices. First: what kind of economy to use. But neither country prospered as much as the world's leading economies. So yes, people earned about as much as their neighbors did, but that wasn't very much. How do they do it? One way is with taxes. Personal income taxes in most countries are progressive: the more money you make, the higher your tax rate. And the more progressive your tax system, the more it reduces inequality. So, for example, while pre-tax income inequality in France is roughly the same as it is in the US, post-tax inequality in France is roughly 20% lower. Meanwhile, inheritance taxes can reduce the amount of wealth that a single family can amass over generations. The US, on the other hand, lets you inherit \$12 million without paying any federal tax. Another way is with transfers - when the government takes tax revenues from one group of people and gives it to another. In Italy, about a quarter of Italians' disposable household income comes from government transfers. A third way is to ensure that everyone has access to things like education and healthcare. A highly educated, healthy workforce can command a higher salary on the market, thus reducing inequality. The fourth way is addressing the digital divide: the gap between those who have access to the Internet and those who do not. A fifth way is dealing with extreme wealth. Multibillionaires can buy social media platforms, news outlets, policy think-tanks, perhaps even politicians, and bend them to their will, threatening the very fabric of democracy. We are just barely scratching the surface of inequality here. We haven't touched on the drastic divides in who has wealth and who doesn't; the power structures that prevent social and economic mobility; and the drastic inequality between countries - the fact that, for example, just three Americans have 90 billion more dollars than Egypt, a country of 100 million people. And here's one final thing to think about: power and wealth are self-reinforcing, which means that equality is not. Left to their own devices, societies tend toward inequality - unless we weaken the feedback loops of wealth and power concentration.	1. The area of this shape multiplied by 2 is the Gini index. 2. You'd never see this in real life because everyone except that one person would starve. 3. But you also never see this in real life, not even in communist countries, because for one thing, that would mean paying everyone - no matter how young, old, what job they're in or where they work - the exact same wage. Typical after-tax Gini in developed countries today are around 0.3, though there's a wide range from pretty equal to pretty unequal. Before we go any further, you should know what the Gini index or any other measure of economic inequality - doesn't tell us: it gives no information about how income and wealth are distributed across genders, races, educational backgrounds or other demographics; it doesn't tell us how easy or difficult it is to escape poverty. And it also gives no insight as to how a particular society arrived at its present level of inequality. 4. In the 20th century, some countries switched to socialism or communism for a variety of reasons, including reducing economic inequality. These changes did dramatically reduce economic inequality in the two largest non-capitalist economies, China and the Soviet Union - especially in the Soviet Union. 5. This and many other issues contributed to the Soviet Union's collapse in 1991. And China, to grow more quickly, shifted its economy towards capitalism starting in the late 1970s. What about capitalist countries? 6. It's tempting to think "no, because the whole point of capitalism is to hoard enough gold coins to be able to dive into them like Scrooge McDuck." China seems to provide the textbook example of this: after it became more capitalist, its Gini index shot up from under 0.4 to over 0.55. Meanwhile, its per capita yearly income jumped from the rough equivalent of \$1,500 to over \$13,000. But there are many counter-examples: capitalist countries in which inequality is actually holding steady or decreasing. France has kept its Gini index below 0.32 since 1979. Ireland's Gini has been trending mostly downward since 1995. The Netherlands and Denmark have kept theirs below 0.28 since the 1980s. 7. Germany and many other European countries have inheritance or estate taxes that kick in at a few thousand to a few hundred thousand Euros, depending on who's inheriting. 8. For example, Social Security programs tax people who work and use the revenue to support retirees.

Table 1: Filtered text and removed sentences from a selected sample (Video 1)

As shown in the sample, this approach reduced some introductory text. For example, the first to the third removed paragraphs are the introduction to the term Gini Index. Although the fourth to sixth removed paragraphs related to ideologies, they were removed since the cosine similarity computes the vectors between the whole word set containing four words. These paragraphs illustrate the difference between capitalism and socialism and do not impact narrative divergence analysis. Similarly, the seventh and eighth remove sentences were removed despite the fact they related to policies. Importantly, the filtered text retains expressions relevant to the narrative. For example, the suggestions to reduce social inequality and critique policies and millionaires remained. This method effectively removes noises in the video texts that are irrelevant to the topic.

However, there is still a pressing problem that needs to be addressed. The videos are pre-selected using the keyword income inequality; the video text could be highly linked to this topic and contain many words like income, wealth, inequality, and gap. Meanwhile, since the video texts were further filtered in the last step through the inequality word set, this problem could be exacerbated.

BERTopic uses the creator's customized class-based term frequency-inverse document frequency (C-TF-IDF). Compared with the classic term frequency-inverse document frequency (TF-IDF), the C-TF-IDF replaced the inverse document frequency with inverse class frequency (Grootendorst 2022). It is based on the assumption that all the documents are made of different classes and all the words belong to these classes. Therefore, it indicates the direct correlation between words and topics. However, this does not correspond to the purpose of the research question. As stated above, the research question is to explore the narrative split between different videos. Critical is the relevance of some particular words for a video, not for a class. Thus, I used the classic TF-IDF, which works efficiently when comparing document differences. The TF-IDF calculates term and inverse document frequency (Jones 1972). The TF-IDF value increases proportionally with the number of the word in a document but decreases inversely with the frequency of the word in other documents.

TF-IDF enables precise control over the frequency of words included in the document analysis. I set the minimum number of appearances to 3 and the maximum appearance percentage to 0.95, which means 95%. With this setup, the words computed in the vectors only include words that appear in at least 3 video texts and exclude words that appear in more than 95% of the video texts, as these words are repeated in most video texts. In other words, We already know these words since these videos are related to the topic of social inequality, and thus, we need to focus on these unique

words in each video. Meanwhile, I also used stop words since they do not have a real meaning for the research question.

In summary, the method employed in this section, which integrates the BERTopic with the TF-IDF, provides substantial advantages for exploring narrative divergence within video content. Leveraging the transformer language model in BERTopic affords a deep contextual understanding of the text, which will retrieve narratives with nuance and precision. This is particularly crucial in handling the complex semantics and varied expressions inherent in video content. In addition, the familiar and unique terms can be identified across videos using the TF-IDF, ensuring that the words considered are only the ones most relevant to defining the narrative divergence. Furthermore, the parameters of TF-IDF help reduce the noise and thus ensure that the common terms are not over-represented, thus enhancing the precision of the whole topic-modeling process.

5.2. Results

This section presents the findings derived from applying the combined BERTopic. BERTopic generated four topics from video content. The first topic, also the greatest topic, is desired as a context topic, which means that BERTopic fails to extract any differences from videos belonging to this topic. This indicates that all of this topic's videos share similar document embeddings. For the research question, this topic represents the widely mainstream narrative and builds a baseline for BERTopic to compare with other topics. The remaining three topics, generated by BERTopic, each represent unique features identified within the video.

Topic modeling results are usually statistically abstract representations consisting of word lists. These words represent possible topics, but the model itself cannot fully understand or explain how these words specifically relate to the true meaning of the text or the contextual environment. People need to name and categorize these topics based on the co-occurring relationships and meanings of these words.

According to the categorized texts in topics, the topics are named and described as follows:

1. Mainstream Narrative: This narrative includes an exhaustive discussion of income inequality, covering multiple perspectives and societal dimensions of impact. Firstly, income inequality restricts individuals' freedom and opportunities, limiting their ability to do what they want due to financial constraints. Poverty and the inability to meet basic needs are directly linked to income inequality. In addition, income inequality hinders access to necessary services for a comfortable life.

It is problematic when the underclass cannot afford basic necessities or access the same services as the upper class, leading to disparities in healthcare, education, housing, and nutrition. Furthermore, unions historically played a role in advocating for worker rights and improving conditions. Government interventions, such as labor laws and regulations, are essential in addressing income disparities and ensuring fair wages and working conditions. This topic also discussed other social factors which impact income inequality in the United States. Race and gender play a significant role in income inequality, contributing to the wage gap between black and female workers. Historical racism and sexism have influenced today's income gap, and there is a need to reduce the racial and gender wage gap by improving the quality of jobs and providing equal opportunities.

2. Wealth accumulation: This topic focuses on how wealth is accumulated and its impact on society. It highlights a key claim: since wealth accumulation is much faster than economic growth, wealth must be concentrated among a few people, which leads to inequality. Meanwhile, such concentration of wealth means that only a few people have significant influence over political decisions, which can lead to policies that favor the wealthy. There is, therefore, a need to ensure that the benefits of economic growth are more widely shared and to avoid a widening gap between the rich and the poor. Adopting policies that aim to distribute wealth more equitably is essential for a fair society.

3. Broken capitalist system: This topic critiques the current capitalist system for its emphasis on maximizing short-term shareholder profits, which is seen as a factor that contributes to economic disparities. Companies have shifted their focus from reinvesting profits in their workforce and equipment to maximizing shareholder returns and increasing income inequality. This topic calls for increased government intervention, including higher wages and greater investment in education, health care, and child care. Most importantly, empowering labor unions through legislation and tax incentives to get the broken capitalist system back on the right road.

4. Bad government intervention: This topic expresses a strong opposition to government intervention. Firstly, interventions reduce the country's economic competition. For example, enterprises close or relocate to countries with less stringent legislation. In addition, raising tax rates on the wealthy reduces their incentive to earn more income since most of it will be taxed. Increasing benefits for low-income groups can lead to the poverty trap, whereby people lose the incentive to raise their incomes because of benefits, as increased incomes may lead to reduced benefits. Hence,

the expression that government intervention has made things worse has appeared several times on this topic.

Table 2 shows the details of each topic. These findings show the diversity of YouTube video content and support the hypothesis that narrative conflicts over income inequality also exist on social media. Each topic not only reflects a unique standpoint within the discourse but also intersects with broader socio-economic theories and existing research.

Table 2: Results of Topic modeling⁵

Topics	Representative words	Related video counts	Selected sentences
Mainstream narrative	mobility, unions, population, debt, earn, stock, richer, black, productivity, uh	174	So long as the services to life are gate kept behind access to wealth, we need to make sure everyone has access to wealth and in terms of how feasible that is, the amount of income inequality we have right now is at record levels.
Wealth concentration	population, assets, households, concentration, europe, ordinary, return, becoming, prices, modern	38	In the long run, there is a tendency for the rate of return of capital to exceed the economy's growth rate, and this tends to lead to high concentration of wealth.
Broken capitalism	shareholders, john, theyve, profits, 40, election, leading, upper, enormous, exactly	18	Now there's a pretty simple explanation for this recent Divergence: corporations no longer share their prosperity with their workers.

⁵ More detailed results can be found in the supplemental materials - Topics_data.csv

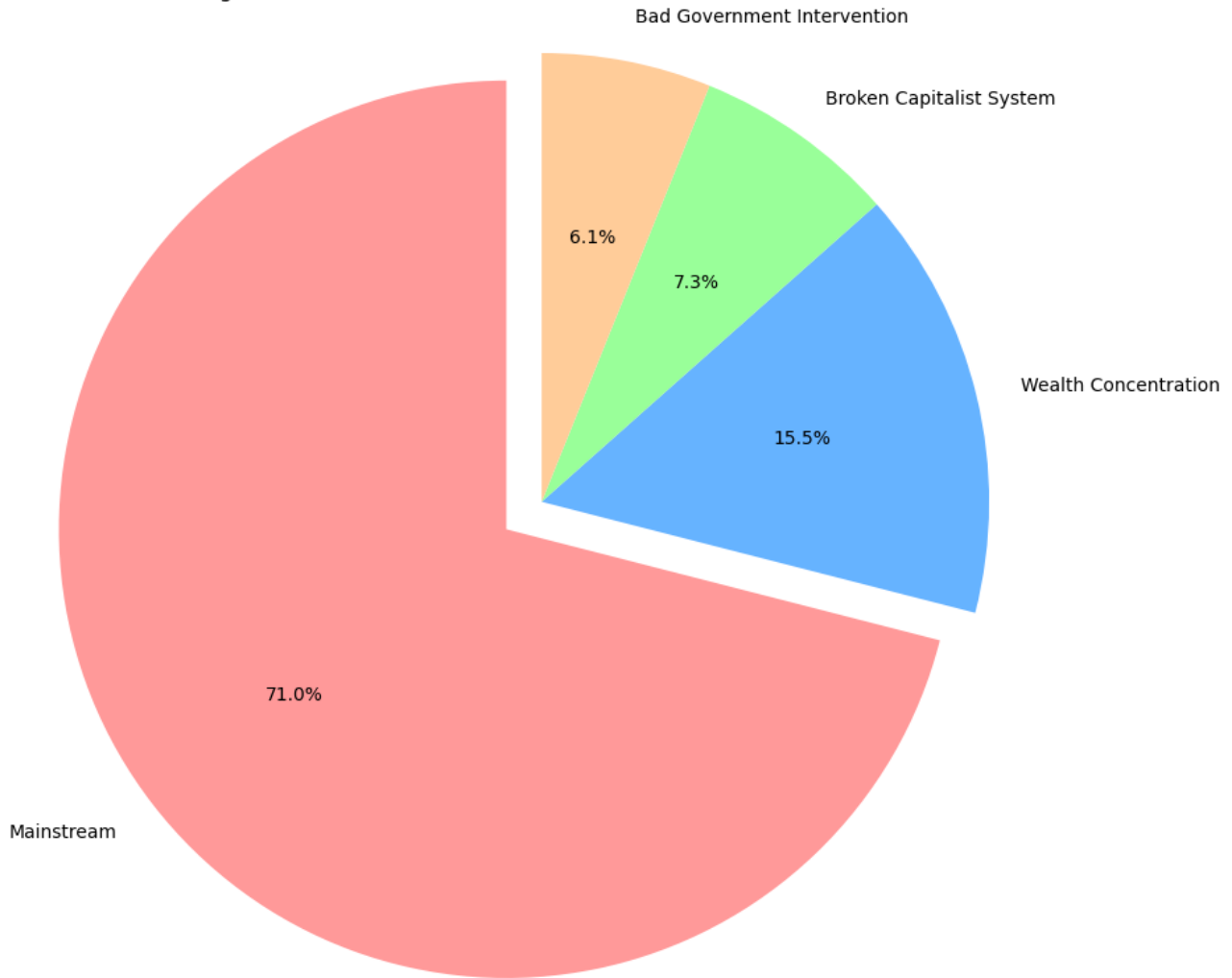
Topics	Representative words	Related video counts	Selected sentences
Bad government intervention	left, trade, dream, candidates, impact, progressive, warfare, failure, candidate, sometimes	15	The enforcement of them needs to be strong if they are actually going to have an impact, but the significant risk of government failure. How businesses May shut down because of how costly these legislations might be, they may relocate to other countries where legislations are not as strict. These are all unintended consequences of government policy here of government intervention, and that can lead to a government failure.

For example, the second topic, wealth concentration, echoes the critiques from Piketty (2014) and Case and Deacon (2021), who discuss the disproportionate influence of the elite on economic policy and the resultant exacerbation of wealth disparities. This narrative on YouTube suggests a growing awareness and critique of the mechanisms through which economic power is concentrated and maintained.

The third and fourth topics are interesting, as they express two opposing arguments. Aligned with Stiglitz (2012) and Mishel and Davis (2015), the third topic characterizes a strong anti-capitalist narrative. This anti-capitalist narrative emphasizes the disconnect between wage growth and profit growth, underscoring the long-term stagnation of wages for the majority. In contrast, profits and incentives for top earners have surged. This perspective is highly critical of the mechanisms within capitalist systems that favor capital over labor, leading to increased economic disparity. Conversely, the fourth one, representing the stand of neoliberal economists such as Hayek (1969) and Friedman (1963), puts forward a strong position in favor of having minimal government involvement. It is convinced that market-driven approaches are more efficient and, in turn, are for the good of society. This narrative views most government policies in reaction to inequality as overreaching and backfiring in their objectives, which may repress economic freedom and innovation. According to this view, the market, insofar as it is relatively left unrestrained, inherently regulates itself and leads to optimal results for everyone, although apparent inequalities may be evident in the short run.

Despite the narrative differences demonstrated between the different topics, in terms of narrative distribution, the majority of the videos still maintain a diverse narrative structure. Of all the 245 videos, 71% are classified as mainstream, which means BERTopic did not capture differences in document embeddings. It is worth noting that being categorized as mainstream does not mean the

Figure 6: Narrative Distribution on YouTube Video Contents



video does not contain content on the other three topics. The mainstream topic also contains some sentences that are against government intervention or higher taxes. However, since BERTopic calculates document embeddings for the entire video content, and these sentences occupy a small share of the overall video content, the videos containing such sentences are not categorized into other topics. It also illustrates the diversity of perspectives still discussed in most YouTube videos on income inequality. The second topic is a scientific, in-depth explanation of the causes of income inequality. It is important to emphasize that about 28.5% of videos focus on a singular viewpoint. This finding exposes a potential risk. If people only look at the videos with one-sided views, they might develop a skewed or incomplete understanding of the complex issues surrounding income inequality. Such incomplete understanding can limit the scope of public debate and reduce the quality of democratic discourse, as citizens may not be equipped with a comprehensive understanding of the issues at hand. This lack of comprehensive perspective can lead to

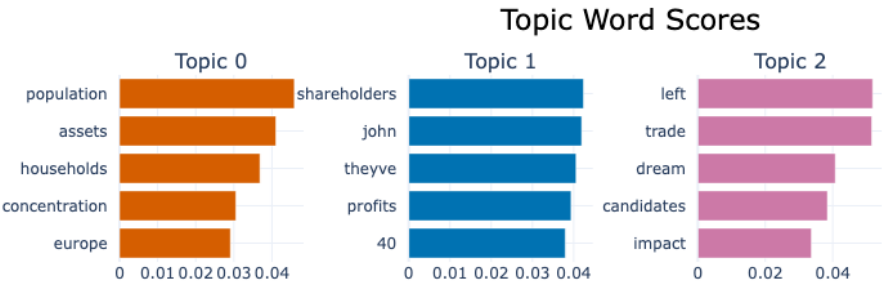
polarization, where individuals become entrenched in their views, unwilling or unable to consider alternative perspectives or evidence. A well-functioning democracy relies heavily on an informed electorate that can understand the nuances of policy issues and contribute meaningfully to public discourse. However, when media consumption is dominated by one-sided narratives, it undermines the diversity of opinions that is fundamental to democratic deliberation. Furthermore, Individuals may be more likely to be influenced by one-sided information to draw erroneous conclusions.

However, this finding does not estimate the impact of algorithmic recommendations. Some groups may be more susceptible to videos containing one-sided views. The importance of the findings urges further research on YouTube recommendation algorithms. So far, there is still a gap in research on how social media recommends content to users. Future research should aim to dissect the mechanics of these algorithms to understand how they prioritize and suggest content to users, potentially reinforcing or diversifying the narratives to which individuals are exposed.

5.3. Limitations

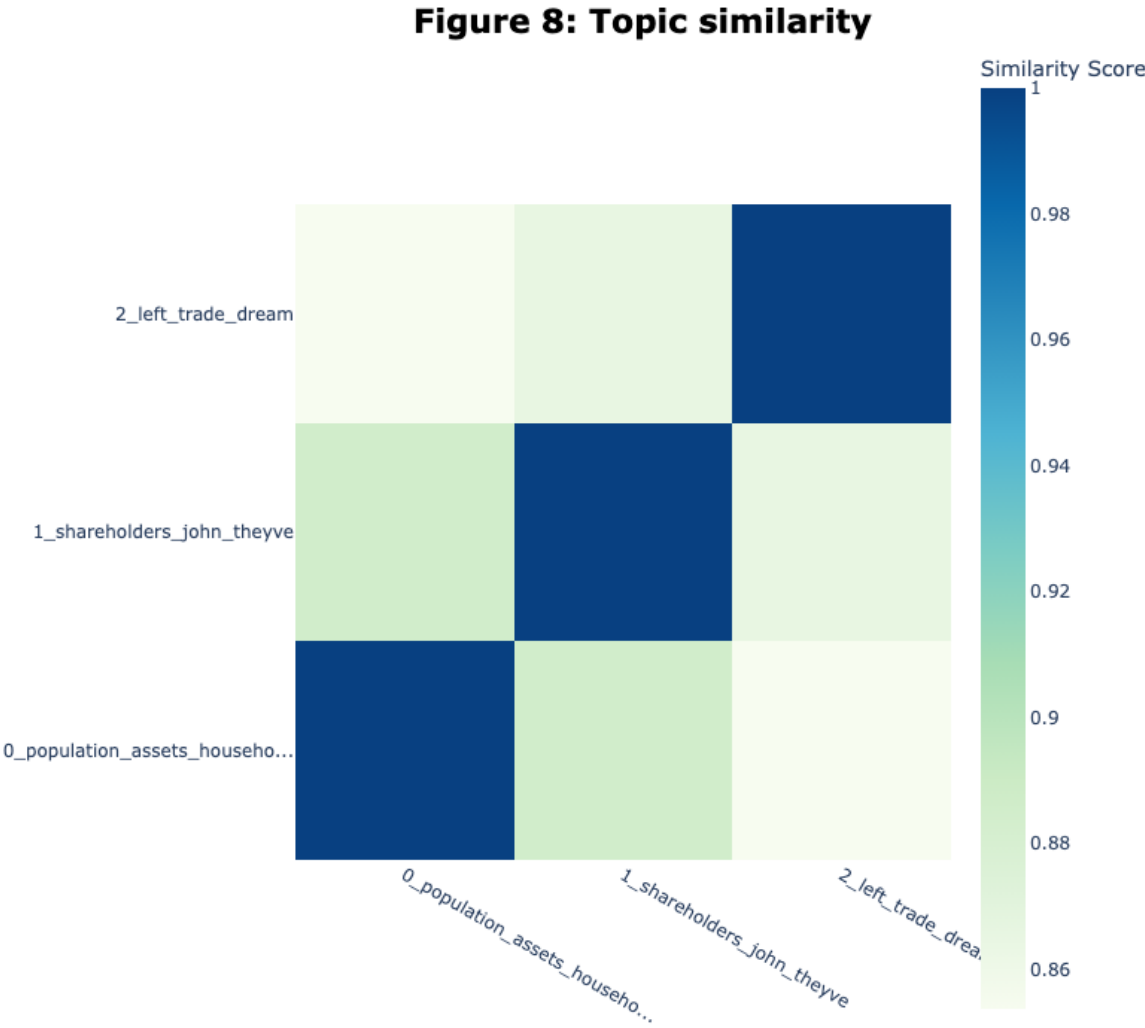
In the analysis, the BERTopic faced significant challenges when differentiating between topics with high semantic similarity. However, the BERTopic effectively identified distinct subset topics within video texts that are highly similar to a common topic, which shows the transformer language model’s strong capability to comprehend contextual nuances.

Figure 7: Topic Word Scores



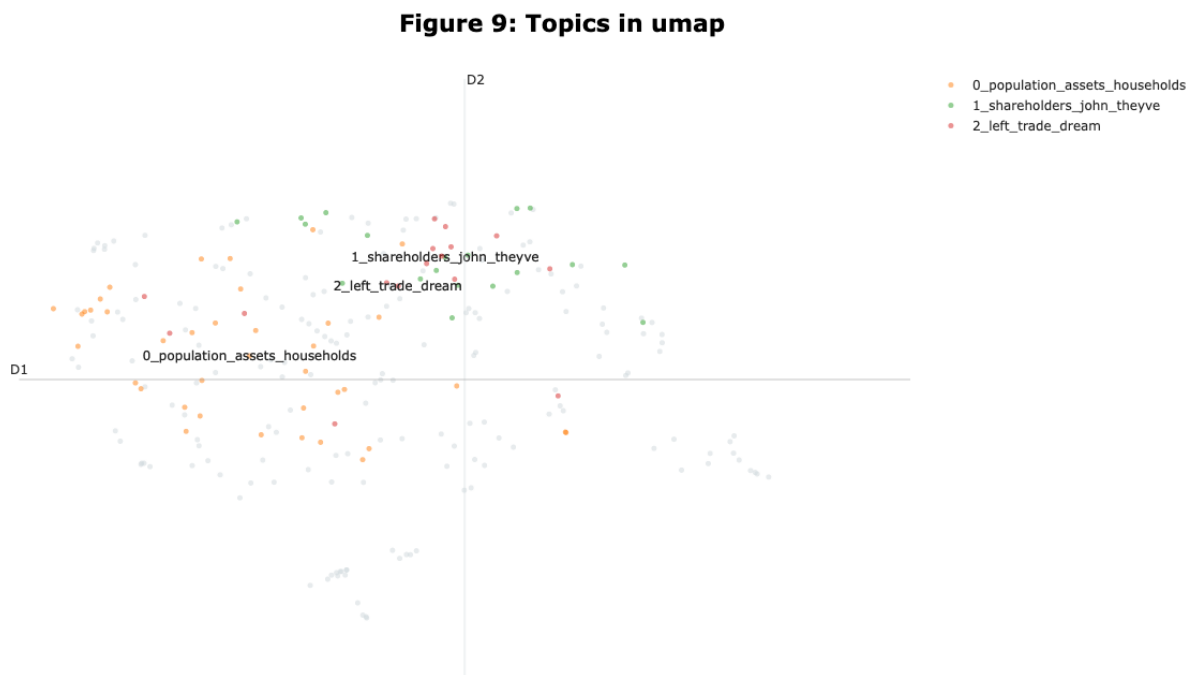
The application of TF-IDF plays a key role in improving the performance of BERTopic. It highlights the importance of individual words in each document. Figure 7 shows the five best representative word scores in three topics, excluding the mainstream topic.

The topic word scores are uniformly low, which suggests two potential explanations. Firstly, although these words occur frequently across the dataset, their presence in multiple videos reduces their distinctiveness, resulting in a high inverse document frequency that diminishes their individual word scores. Alternatively, these words might inherently have a low frequency of occurrence combined with a low inverse document frequency, contributing to their low scores. If it is the first reason, then the validity of topic modeling is questionable since the best representative words occur in most videos. Since the threshold in TF-IDF filtered words appears in more than 95% of the videos, it is unlikely that the low topic word scores are the result of a high inverse document frequency. The low topic word scores actually indicate a successful aspect of the topic modeling process. TF-IDF successfully distinguished between topics by recognizing rare words.



The Figure 8 illustrates the high semantic similarity among topics. Semantic similarity between all three topics is from 0.85 to 0.88. This suggests that these topics discuss variations of similar themes or share a common conceptual framework. In the analysis, all the topics related the common topic income inequality. Despite the challenges inherent in distinguishing closely related topics, the visualization demonstrates that the BERTopic method effectively captures the subtle differences and overlaps between them.

Figure 9 presents a Uniform Manifold Approximation and Projection (UMAP) visualization. In the graph, each point represents a document embedded in two-dimensional space. According to the color, points belong to different topics. Gray represents the first context topic, orange represents the second topic (Wealth concentration), green represents the third topic (Broken capitalism system) and red represents the fourth topic (Bad government intervention). Due to the high semantic similarities, the identified three topics are not well isolated. As shown in Figure 9, some points from different topics are overlapped.



Although the modeling processes are very successful, there might be some limitations. Firstly, the transformer language models are pre-trained on universal corpora like newspapers and blog posts. These models are not specifically trained for politics-related texts. When filtering irrelevant text with all-mpnet-base-v2, some sentences remain in the text even though they are clearly

irrelevant to the two word sets. Similarly, there may be some relevant sentences that have been removed by mistake. This may cause BERTopic to be biased when calculating document embeddings. In addition, as shown in Figure 8, the transformer language model reached its limit in computing semantic similarity. That is, the model has reached the maximum capacity to recognize the topics, and thus, there might be some potential topics that are not identified due to the high semantic similarity. In future research, models trained specifically on political texts may produce more accurate results. In addition, effective filtering of text remains critical. Improved preprocessing steps, such as more accurate word sets or enhanced sentence embedding techniques, could reduce the errors that occur in filtering.

6. Analysis of consumption: The polarization of interaction

Emotional polarization, as defined by Baldassarri and Page (2021), creates strong in-group and out-group boundaries, leading to varied attitudes towards members of one's own group compared to those of opposing groups. This phenomenon is evident on platforms like Facebook and Twitter, where users tend to interact with others who share similar political views. However, it is unclear if the same level of polarization is present on YouTube. This chapter aims to answer the following question: How polarized are the interaction networks of videos about income inequality on YouTube? Based on existing findings on Twitter and Facebook, it is hypothesized that the comments on YouTube show clear homogenization, with users interacting mostly with others who share their political stance.

This chapter used a fine-tuned BERT model to classify the comments' political affiliation and calculated the Krackhardt E-I index (Krackhardt and Stern 1988) to explore the homogeneity of the interaction network. The methodology section explains why this method was selected. The results section then analyzes the outcomes and discusses potential directions for future research.

6.1. Methodology

In order to explore the political stance of comments, it first needs to categorize the text. Text classification includes two methods: traditional machine learning and deep learning. Machine learning is based on the principle of feature extraction through predefined rules (algorithms) on known data and to predict unseen data through learned patterns (Bishop and Nasrabadi 2006). Therefore, the core of the machine learning method is the algorithm, by text classification, the text preprocessing rule, and the classifier. Text preprocessing determines what features in the text will be extracted. There are mainly three ways to extract textual features, such as bag of words, TF-IDF, and word embeddings. The bag of words calculates the frequency of each word in documents. The TF-IDF further considers the importance of words for one document to reduce the impact of common words. Word embeddings, however, provide a dense representation where each word is represented by a vector in a continuous vector space. Unlike bag of words and TF-IDF, which primarily focus on the presence and frequency of words, word embeddings consider the actual usage and context of words within a text through their relative position and neighbor words. Classifiers are the second crucial step of the machine learning method since they determine the

learning performance. Five commonly used classifiers are Support Vector Machines, K-nearest neighbors, Naive Bayes, Decision Trees, and Random Forests (Shalev-Schwartz and Ben-David 2014). Support Vector Machine aims to find the best hyperplane that provides the maximum margin between the closest points from each class. The mechanism of the Support Vector Machine is divided into two steps. First, the extracted features in the data are vectorized, and then the optimal hyperplane is found to maximize the minimum distance between different classes. The K-Nearest Neighbor aims to predict by finding the closest K neighbors in features. The key to this approach is the comparison of the training set and the data to be predicted. K-Nearest Neighbor measures the distance between points in the two datasets and classifies them as nearest. In this way, this algorithm can classify points from data to be predicted into the most similar (i.e. nearest neighbor) category. Naive Bayes is based on Bayes' theorem and assumes that features from data are independent. This classifier computes the probability that each feature in training data belongs to each class. The features are then classified according to the maximum probability. Decision Tree and Random Forest are two similar classifiers. Decision Trees is based on the tree structure and follows the if-then-else rule. At each tree node, a decision is made to split the training data into smaller subsets based on certain conditions. This continues until the subsets are homogeneous enough to make a classification decision or until they meet a stopping criterion, such as a maximum depth of the tree. On the other hand, Random Forest improves upon the Decision Tree methodology by creating an ensemble of random Decision Trees. Each tree in the forest makes a vote for the classification, and the class receiving the majority of votes becomes the model's prediction. This approach significantly reduces the risk of overfitting and usually results in better generalization on unseen data compared to a single Decision Tree.

However, in this study, the machine learning method faces two challenges due to the characteristics of social media text. First, texts on social media are non-standardized. Short texts have only a few to a dozen words and lack sufficient word co-occurrence or shared context. Meanwhile, these texts could contain many misspellings and noise. Therefore, it is tough to extract effective linguistic features from such texts (Song et al. 2014). The classifiers from machine learning methods like Support Vector Machine and Naive Bayes compute in vector space. The inaccurate feature vectors make the classifiers unable to distinguish between different categories. According to Zhou et al. (2010), K-Nearest Neighbors and Naive Bayes have low performance on short texts. The second problem is the generalization. Generalization means the performance of

classifiers that work on untrained data. Due to the mechanism of the machine learning method, the training data should contain as many features as possible for the classifiers to learn. However, texts on social media come from various users, all of whom have their own language styles. In other words, these data are high-dimensional and nonlinear. Even expansive training datasets may not capture the full spectrum of linguistic features in social media texts. Consequently, when classifiers from machine learning encounter features not represented in the training set, they are prone to misclassification.

The deep learning method offers a better solution than the traditional machine learning method. Its greatest advantage is its ability to learn based on artificial neural networks. According to Kohonen's definition (1988), artificial neural networks are large networks made up of simple, often adjustable elements (Neuron) and hierarchical organizations. They are designed to respond to things in a way similar to the human brain. The neuron and hierarchical organizations allow neural networks to compute information in parallel and in multiple layers. As data moves deeper into the network, each layer refines and builds upon the features recognized by the previous layer, allowing the identification of increasingly complex features. The second advantage is the performance on generalization (Haykin 2010). Generalization refers to the model's ability to perform well on new, unseen data without overfitting on learned features. Despite ongoing research, the reasons behind the strong generalization performance of deep learning, even on noisy data, remain not fully understood (Zhang et al. 2021; Chatterjee and Zielinski 2022; Kawaguchi et al. 2017). One explanation is that the multilayer network of neural networks produces an effect similar to Occam's razor (Midgard et al. 2023). Valle-Perez et al. (2018) introduce the idea of a simplicity bias in deep learning, where networks are inclined to learn simpler patterns first, which may explain their effectiveness with high-dimensional data. In essence, each layer outputs the simplest interpretation of the incoming data, contributing to the model's overall ability to generalize from training data to real-world applications.

Depending on the network structure, there are three approaches from the deep learning method for text classification: Recurrent Neural Network (RNN), Convolutional Neural Network (CNN), and Transformer models. The input-output processes are circular on RNN. It uses the same parameters for each time step and takes in both the current input and the previous output. This approach helps RNN consider past information when making current decisions. For text classification, RNN turns sentences into vectors by pre-training (Hu et al. 2020). Then, these

vectors were used to encode sentences and predict their categories. However, RNN often has problems with vanishing or exploding gradients when they take in a lot of long sequential information. To fix these issues, Long Short-Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997), which adds a forget gate, was developed. CNN was first used primarily to process image information. CNN's input layers have convolving filters that are applied to capture features. Kim (2014) was the first to use convolutional neural networks for text classification. In his study, sentences are turned into matrices, with each row being a word vector from the sentence. Convolutional filters of various sizes slide over the matrix to process the data, and the outputs go through a max-pooling layer. However, CNN has limitations with long sentences due to the fixed size of the sliding window in convolution.

Both RNN and CNN have their limitations. Since RNN relies on one-way sequential inputs, it needs to improve in understanding the context. For example, social media users tend to be very casual and informal in their oral expression, and critical information may appear in any position. Since RNNs process text in a front-to-back order, they may miss or misinterpret important information only elucidated later in the text. CNN, despite its advantages in extracting local features, however, the filter size in CNN restricts its ability to analyze distant connections within the data, especially when dealing with long sentences with many clauses.

Compared with RNN and CNN, Transformer models introduce a new mechanism called attention (Vaswani et al. 2017). It assumes that information is categorized as important and unimportant and that the important part needs to be attended to more. The attention mechanism parallel computes each word's query, keys, and values vectors (Vaswani et al. 2017) and uses these vectors to calculate the attention score for each word. The attention score is the dynamic relevance of words in the whole input, or in other words, how important a word is for another part. The attention score's size determines a word's weight in the final output. In contrast to the sequential input of RNN and local feature extraction of CNN, Transformer models compute the features of all inputs simultaneously and do not depend on previous outputs. Due to the advantages of the underlying logic, Transformer models outperform RNN and CNN in natural language processing tasks (Tang et al. 2018).

Bidirectional Encoder Representations from Transformers (BERT) and its variants are the current best performing Transformer model for text classification (Devlin et al. 2018). Compared with default Transformer models, it contains bidirectional encoders, which allow each word in the text to

have its representation computed, taking into account both the contextual information on its front and rear sides. In addition, BERT is pre-trained on large corpora using two tasks, which are the Masked Language Model (MLM) and Next Sentence Prediction (NSP). These tasks are designed to provide a comprehensive understanding of language context and structure in advance. The key step in the application of BERT to text categorization is fine-tuning. Fine-tuning always improves task performance (Zhou and Srikumar 2021; Hao et al. 2019). The study found that fine-tuning can increase the distances between examples associated with different labels, leading to better generalization and improved performance on downstream tasks (Zhou and Srikumar 2021). Additionally, fine-tuning can enlarge the gaps between label clusters, allowing for better separation of classes and potentially improving classification accuracy (Zhou and Srikumar 2021).

In this study, a total of 344,227 comments need to be annotated. Considering that the size of the training set has a significant effect on the effectiveness of fine-tuning (Zhang et al. 2020), the training set was set at 4,000 randomly selected samples. Due to the long-tailed distribution of the comments (see Figure 5), the samples were mainly derived from a few videos with a high amount of comments. Given the hypothesis that individuals are likely to respond to comments reflecting similar political views, it is theorized that videos with fewer comments may exhibit a higher concentration of comments from the same political affiliation. Increasing the representation of videos with fewer comments in the training set may disrupt category balance. Thus, no intervention was made to adjust the random sampling.

Table 3: Distribution of sample origins

Video index	10	47	227	42	91
Sample count	773	689	251	228	166

In this study, social media comments were manually annotated to examine public attitudes towards inequality and the capitalist system. Three distinct labels were applied based on the content of the comments:

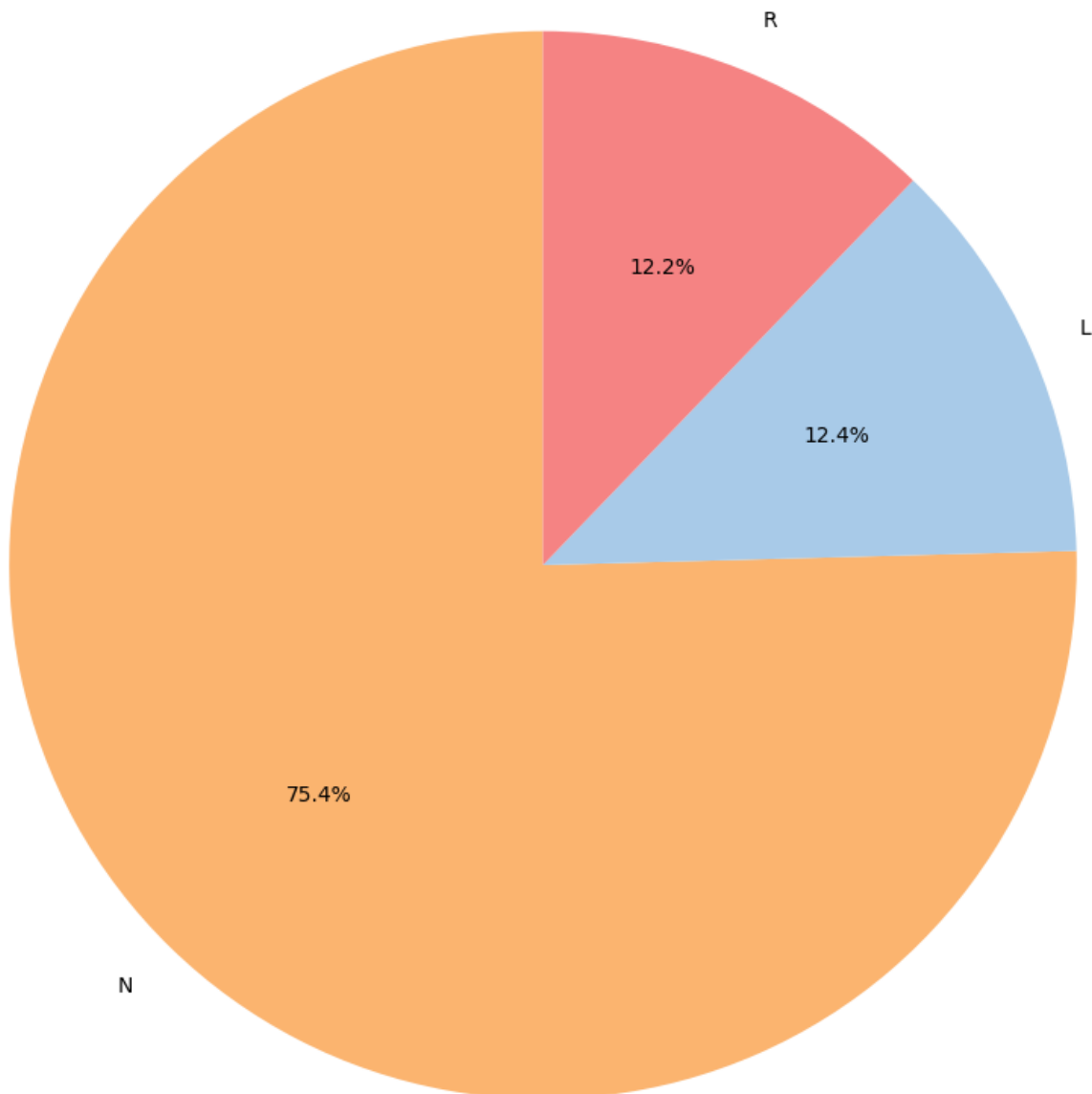
Label L: Assigned to comments explicitly expressing dissatisfaction with inequality and the capitalist system and advocating for more government intervention, such as redistributive taxation policies.

Label R: Applied to comments that justify inequality by attributing it to individual effort or talent differences and explicitly oppose government intervention.

Label N: Used for comments that either do not clearly express an attitude towards inequality or are ambiguous in their semantics.

A significant challenge in this labeling process was the diverse nature of social media texts. Many comments, while not explicitly discussing inequality, expressed dissatisfaction with specific political parties or contained partisan hate speech. In cases where such opposition was related to policies with ideology implemented by these parties, such as market liberalization, I assigned a label of L if the comment opposed these policies and R otherwise. Comments that were purely verbal attacks without any clear policy context were labeled as H. This is because it is difficult to infer the commentator's position on inequality through these comments.

Figure 10: Label Distribution in the Training Set



However, in the ensuing fine-tuning process, the label H is not considered separately. This is because this study aims to assess the homogeneity of attitudes towards inequality within social media interaction networks. Including hate speech as a separate category was deemed not to align with the objectives of this analysis. In addition, a larger category would increase the complexity and difficulty of fine-tuning the BERT model. Considering the above, all comments labeled as H were merged into the N label.

The labeled training set contains 3,017 N labels, 495 L labels, and 488 R labels. The distribution of labels within the training set is shown in Figure 10. The distribution highlights a significant prevalence of neutral (N) comments, totaling 3,017, compared to 495 comments advocating for more government intervention (L) and 488 justifying inequality (R). This imbalance in the distribution should be focused on during the fine-tuning process, and it can lead to model training overfitting on a more representative category (Madabushi et al. 2019).

The next phase involved fine-tuning the BERT model using the labeled training set. Initially, the dataset was divided into two subsets: a training subset and a test subset. The purpose of the test subset, which comprised 20% of the total training set, is to evaluate the model's ability to generalize to unseen data. This division ensures that the performance metrics obtained from the test set reflect how the model might perform in other data outside of the training set. Then, the textual data was processed using a tokenizer designed for the BERT model. This involves transforming the raw text from both the training set and test set into lists and then tokenizing these lists. The tokenizer was configured to return data in NumPy array format and to apply padding to ensure that all sequences have the same length, a requirement for batch processing in neural networks. Alongside the tokenization of the text, the text labels for both the training and testing datasets were converted into integers and stored as NumPy arrays. These labels, which are critical for supervised learning, correspond to the outputs that the BERT model will predict. After completing the above steps, the textual data was transformed into a form compatible with the BERT model.

The crux of fine-tuning is how to deal with imbalanced classification. Although the Transformer-based BERT model adapts well to imbalanced data compared to other classification methods, if the data is too imbalanced, the BERT model will tend to predict unseen data as the majority of the classification. Three standard solutions to the imbalance problem exist: category weighting, random undersampling, and random oversampling (Johnson and Khoshgoftaar 2019). Category weighting involves adjusting the weights of the classes during the loss computation phase. This approach

gives more weight to the minority class, directing the model to pay more attention to these under-represented classes during training. It does not alter the actual data distribution but modifies the learning process to focus more on minimizing errors in the minority class. Random undersampling and random oversampling are approaches that change the amount of data. While random undersampling reduces the size of the majority class by randomly removing samples, random oversampling involves duplicating examples in the minority class to balance the class distribution. Through many attempts, the oversampling approach led to a better accuracy of the BERT model.

Several key hyperparameters need to be carefully selected and tuned to implement these strategies and optimize the BERT model. These hyperparameters include the optimizer, the learning rate, the loss function, the batch size, and the number of epochs for fine-tuning. In tuning these hyperparameters, I considered two important indicators: the model’s accuracy on the training subset and the loss on the testing subset. Furthermore, the memory of the Graphics Processing Unit (GPU) is also an essential factor in the selection of hyperparameters such as batch size and loss function. Through several attempts, the learning rate was set to $5e - 6$. This is a minimal value to avoid early overfitting of the model. I used the Adam optimizer due to its adaptive learning rate capabilities. For the loss function, I selected the sparse categorical cross-entropy loss, which is appropriate for multi-class classification problems where the labels are provided as integers. The batch size was set to 8 due to the constraints of GPU memory. To determine the optimal epoch times, I implemented early stopping. This technique monitors the validation loss during training and stops the training process once the validation loss stops improving. The specific settings for early stopping included monitoring the validation loss, with a minimum delta of 0.1 and patience of 0 epochs, meaning that training stops as soon as the validation loss fails to improve. With early stopping, the optimal epoch time is set to 2. The accuracy on the training subset is 0.83, and the loss on the testing subset is 0.66.

Table 4 below illustrates the performance of the fine-tuned BERT model on the test subset, displaying the precision, recall, and F1 score for each label and the overall model. The F1 score is the key metric for evaluating the model’s performance. The table reveals that the model has a low F1 score for the R label. This low F1 score for the R label is primarily due to the model’s tendency to incorrectly predict non-R-labeled posts as R-labeled, as indicated by the precision and recall values. N label has, in general, a better performance than L and R labels. Its F1 score reached 0.89.

Table 4: Summary of the precision, recall, and F1-score for each label

Label	Precision	Recall	F1 score
N	0.91	0.86	0.89
R	0.36	0.58	0.43
L	0.55	0.52	0.53
Weighted average	0.79	0.72	0.76

To better understand where the model is making errors, we need to examine the normalized confusion matrix. Table 5 highlights specific areas of incorrect predictions. The two non-diagonal values in each row represent the proportion of misclassifications for other labels. The model’s main errors occur with the R and L labels. Overall, the model underestimates the number of labels R and labels L. It misclassifies R-labeled comments as N label and L-labeled comments as R label. The performance losses on R and L labels may be due to the imbalanced training set. Although oversampling was used to address the imbalance in the training set, the large ratio between the majority and minority labels resulted in comments with minority labels being duplicated many times. This simple replication did not increase the diversity of the text. Oversampling increases the prediction frequency for minority labels in the fine-tuned model, but it does not necessarily improve accuracy. For example, the R label has a high recall value but a low precision value. Since the fine-tuned BERT model achieves valid predictions on the test subset and an adequate F1-score of 0.76, it was used to predict the political stance of the comments across the whole dataset.

Table 5: Normalized confusion matrix

Label	N	R	L
N	0.86	0.1	0.05
R	0.30	0.53	0.18
L	0.20	0.27	0.52

After accomplishing text classification, the next step is to measure homogeneity in the interaction network. This involves analyzing how similar or diverse the interactions are within different groups based on the classified political stances. To quantify this homogeneity, the Krackhardt E-I index (Krackhardt and Stern 1988: 128) was used. The E-I index defines:

$$\text{E-I Index} = \frac{E - I}{E + I}$$

The E represents the number of all heterogeneous interactions in the network, i.e., the sum of edges of nodes with non-identical political stances. The I represents the number of all homogenous

interactions in the network, i.e., the sum of edges of nodes with the same political stances. Interpreting the E-I index provides insights into the network's homogeneity. An E-I index close to 1 indicates a high number of external connections, suggesting high interaction diversity. Conversely, an E-I index close to -1 indicates a high number of internal connections, suggesting that interactions are more homogenous within groups. In this study, the classification of links as internal or external is determined based on the political position of the comments involved. Specifically, if a secondary comment does not share the same political label as the primary comment to which it is responding, it is counted as an external link. Conversely, if a secondary comment shares the same political label as the primary comment, it is considered an internal link.

In a nutshell, this research applies a detailed methodology to measure the homogeneity of the interaction network and political stance in social media comments. For text classification, the research used a fine-tuned BERT model, making the most of its ability to work with noisy and non-standardized text. Random oversampling was used to address the imbalance in the training set. Key hyperparameters, such as learning rate, optimizer, loss function, batch size, and number of epochs, were carefully tuned to optimize the model's performance. Early stopping is implemented to avoid overfitting and to ensure the generalization of the model to new data. The E-I index formulated by Krackhardt and Stern measures homogeneity in the interaction network and gives insight into the diversity and clustering of political stances, which potentially allows a thorough analysis of social media interactions and their political dynamics.

6.2. Results

Figure 11 illustrates the distribution of labels across all comments on YouTube regarding political positions on inequality. The data reveal that the majority of comments (71.8%) are classified under the label N, indicating that they do not clearly express any political attitudes. In contrast, only 28.3% of comments display a definitive political stance, with 11.9% labeled L for left-leaning views and 16.4% labeled R for right-leaning views. These statistics suggest that YouTube may not be a primary platform for political debates.

Table 6 indicates the homogeneity of all the comments. Label R has fewer internal links (6,228) compared to external links (14,709), resulting in a positive E-I index (0.41). This indicates that the group that justifies inequality is more outward-looking, engaging more with the outside than within itself. Label L similarly shows a higher engagement with external links (9,296) compared to internal

links (4,150), with an E-I index of 0.38, pointing to a tendency to interact more with other groups than within the group against inequality. In general, the comments with political stances (L and R) have a positive E-I index (0.40), indicating that interactive networks are not homogeneous.

Figure 11: Label Distribution in the whole Comments

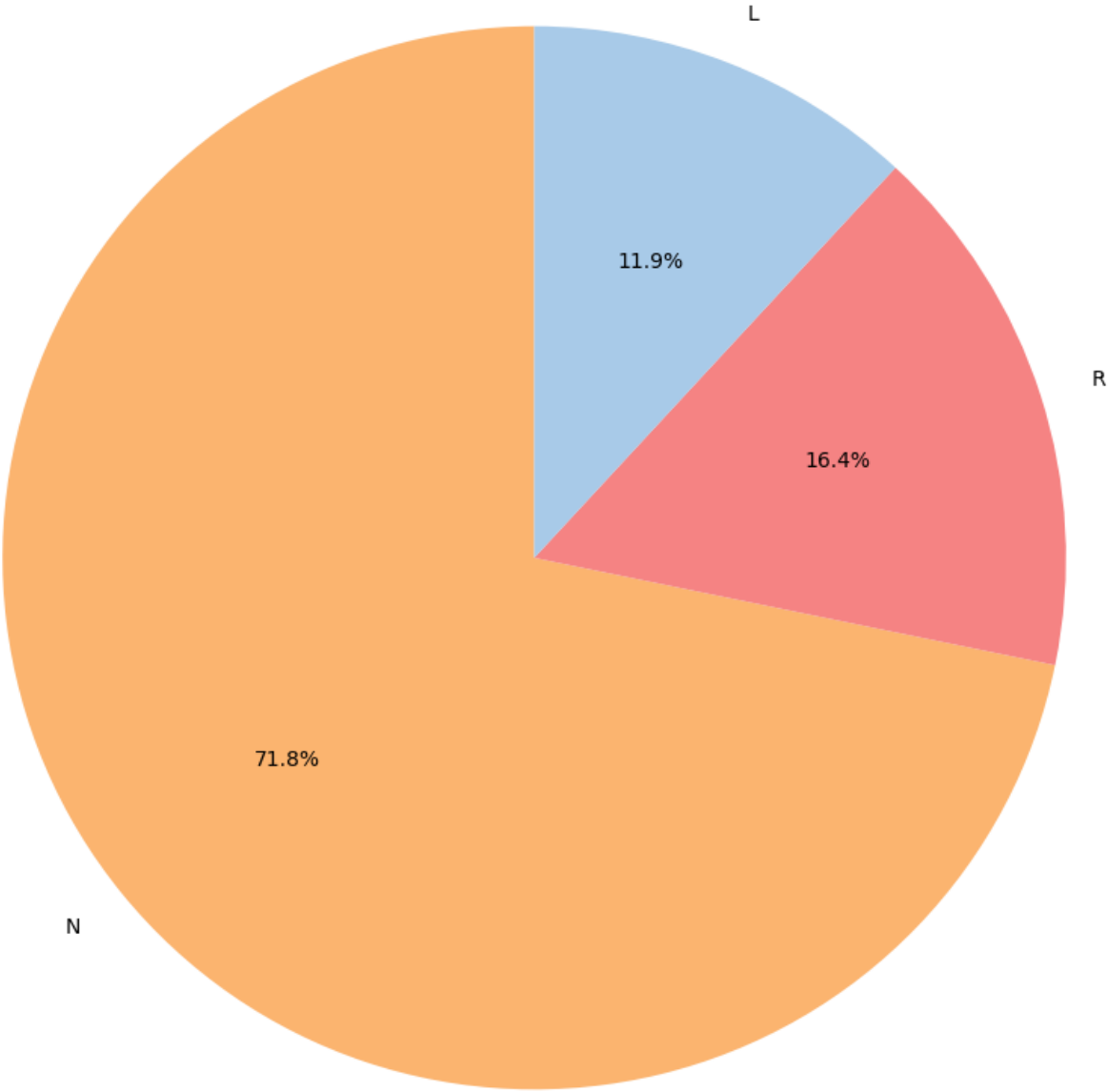
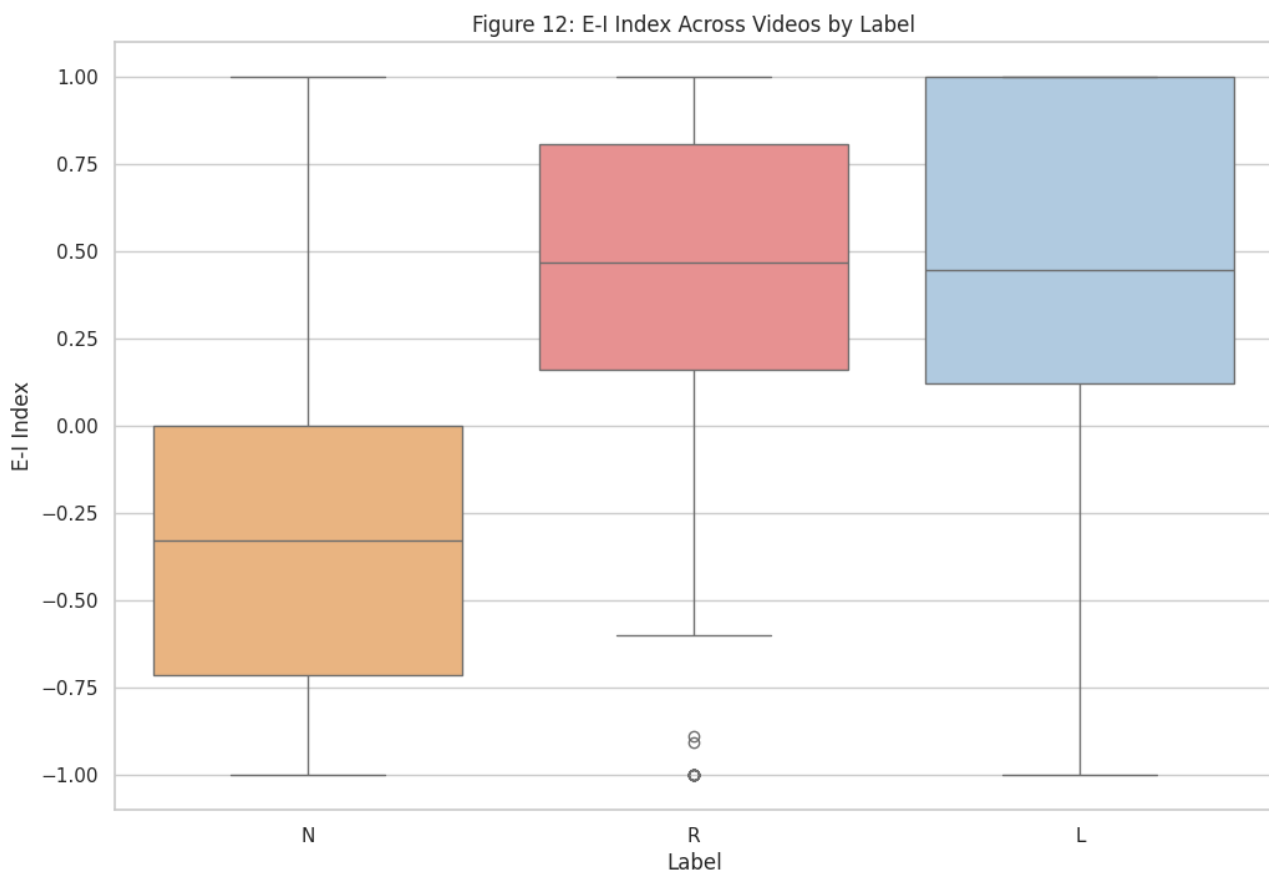


Table 6: The E-I index in the whole comments

Label	Internet links	External links	E-I index
N	70145	35868	-0.323
R	6228	14709	0.41
L	4150	9296	0.38

Sum (R + L)	10378	24005	0.40
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While the global E-I index provides a broad overview of interaction patterns across all videos, it may obscure the homogeneity in interaction networks that manifest within some videos. Figure 12 illustrates the boxplots of the E-I Index for different labels (N, R, L) across videos. Boxes from both R and L labels are above zero, which means that most comments belonging to these two labels tend to interact with other groups. However, it is important to note the presence of outliers in the boxplot for label R, which is close to -1. For label L, the whisker reached -1. These observations suggest that in some videos, comments labeled R and L demonstrate a homogeneous tendency towards internal interactions, contrary to the general trend. This internal focus is particularly evident in videos with fewer comments, where individual interactions can disproportionately influence the overall interaction pattern. Furthermore, there is no evidence that the homogeneity of interaction networks is related to the narrative differences found in the previous chapter.



In conclusion, the overall positive E-I indices across multiple videos (see Figure 12) and the generally positive global E-I index (see Table 6) suggest that on YouTube, users predominantly engage in discussions with individuals holding differing political affiliations. This trend indicates a broader pattern of heterogeneity in user interactions. However, a notable exception occurs in videos

with fewer comments, where evidence of homogenization emerges, showing users interacting more frequently with those who share similar political views.

The results rejected hypothesis 2 in general. In most of the videos related to income inequality, no evidence of interactive polarization was found. This finding contrasts with previous research on Twitter and Facebook, which suggests that people are more likely to interact with others who share the same political affiliation ((Barberá et al. 2015; Rivero 2017; Garimella and Weber 2017; Conover et al. 2011; Barberá 2015; Bakshy et al. 2015). The findings confirm Cinelli et al. (2021)'s argument that group polarization is due, to a greater extent, to the nature of the platform. In contrast to other platforms, the distribution of information on YouTube does not rely on following networks. This can lead to YouTube users being exposed to information from different viewpoints. Moreover, the results do not suggest that social media users avoid interacting with others who hold different opinions. On the contrary, E-I indexes greater than 0.5 overall indicate a preference for debating with people who hold opposing views. This points to a more complex interaction pattern on YouTube, where users are not only exposed to diverse content but also engage with it in ways that may reduce polarization.

This finding challenges the view in the existing literature that social media exacerbates political polarization and highlights the importance of platform design in shaping user interactions. Future research could further focus on how the characteristics of different social media platforms lead to variations in user interactions. Investigating these differences can provide insights into why platforms like YouTube may expose users to diverse viewpoints while others like Twitter and Facebook might reinforce existing beliefs.

Existing research still lacks an understanding of how interactions on social media affect political attitudes in the real world. Karlsen et al. (2017) found that in online debates, participants' original political positions are reinforced, regardless of whether they are exposed to similar or opposing viewpoints. Due to data and methodological limitations, this finding is restricted to users' interactions on YouTube, a single social media platform. Whether the cross-partisan communication observed in this study strengthens or weakens users' existing political attitudes remains unknown. In other words, this study does not address ideological polarization. Future research will need to employ methods that span both social media and real-world interactions to understand these dynamics fully.

6.3. Limitations

This study encountered a limitation due to prediction errors in the fine-tuned BERT model used. As stated before, it demonstrated a bias in the predicted count of labels. Since the E-I index depends on the outputs from the fine-tuned BERT model, the final results might be disturbed by the biased counts.

The prediction errors are particularly noticeable with two specific labels, R and L. As detailed in Table 5, the fine-tuned BERT model underestimates the counts for both labels. Comments that should be labeled as R are sometimes incorrectly predicted as N. In contrast, comments that should be labeled as L are occasionally misclassified as R. There are two potential impacts on the results shown in Table 6. For the label R, being misclassified as N might result in fewer external links and more internal links than exist.

Conversely, for label L, being misclassified as R could result in an overestimation of external links and an underestimation of internal links for label L. Additionally, this misclassification might cause an inaccurate recording of fewer internal links and more external links for label R. However, given that the current results show twice as many external links as internal links for the R and L labels, the final finding - that users are more likely to interact with other people of differing political stances on YouTube - must not be overturned, even when considering these potential prediction errors.

The prediction errors primarily stem from an imbalanced training set, where certain classes are underrepresented. Currently, there is a lack of better academic methods specifically designed to address text diversity in natural language processing (NLP) tasks when faced with insufficient class counts (Johnson and Khoshgoftaar 2019). Further research could improve the shortfall in three ways. Firstly, employing a more extensive training set is an effective strategy. Increasing the size of the training set means that more diverse samples can be included. For the underrepresentative classes, adding samples can help the model to learn better and recognize their features, thus reducing the bias towards the dominant class. Secondly, more advanced language models may be able to solve the imbalance problem in the future. Thirdly, modifications at the output layer could enhance the model's ability to focus on meaningful classifications. In the training set, the label N does not convey any political stance and thus holds no relevance for the purpose of the research. Training the model on the features associated with label N could mislead the prediction for the other two meaningful labels. In future studies, there are bound to be similar problems with meaningless

labels. A practical modification would be to configure the model to exclusively recognize and learn from the features of the two politically relevant labels. Additionally, the prediction thresholds for these labels should be raised to ensure that only the most clearly aligned comments are categorized accordingly. When the model determines that comments do not meet these criteria, they should be automatically classified into a third label as meaningless. With the above approaches, the shortcomings of the present method are expected to be improved in future studies.

7. Analysis of distribution: The echo chamber effect

In the previous chapter, the examination of homogeneity within interaction networks revealed a lack of clear evidence. Moving beyond this finding, this chapter shifts focus to another influential aspect: the impact of video content. Specifically, it addresses the question: How are the political stances expressed in comments associated with the video's content? To explore this relationship further, Hypothesis 3 is proposed: The more potent a video articulates a specific political viewpoint, the greater the likelihood of attracting comments that echo this stance. Confirmed with the echo chamber argument in the literature review, this hypothesis assumes that viewers are drawn to content that resonates with their existing beliefs, thereby influencing the nature of interactions in the comments section.

The methodology section delineates the methods employed and provides rationales for their use. Subsequently, the results section presents the findings and discusses their implications for future research. Lastly, the limitations section critically evaluates the constraints of the findings.

7.1. Methodology

The challenge for the study is measuring the strength of a particular viewpoint in a video. The measurement aims to “instantiate some concept within the hypothesis or theory to facilitate quantification (Grimmer et al. 2022: 275).” Specifically, this study needs a value to directly reflect the strength of each video's viewpoint on income inequality. Three traditional methods of converting text to values are sentiment analysis, word counts, and human rating.

Sentiment analysis is a method that examines subjective information from text data to determine the sentiment or emotional tone expressed within the text. It usually creates sentiment scores in three dimensions: positive, neutral, and negative. The dataset for this study includes videos in various formats, such as podcasts and documentaries. These formats inherently differ in style and consequently influence the emotional dimensions of the content conveyed. Documentaries typically maintain a more neutral tone, while podcasts often display more pronounced negative or positive sentiments. Due to these stylistic differences, relying solely on sentiment analysis to assess the strength of political positions in the videos may not yield accurate results.

Word counts or dictionary approach is a method that quantifies the prevalence of specific words or phrases to relevant topics. This method employs a list of words, each associated with a tone score

based on the topic and their respective frequency rates, to assess the tone of a document (Grimmer et al. 2022: 287). The disadvantage of this approach is that its validity depends on the dictionary settings. The tone score assigned to words must accurately reflect their context-specific usage. However, Barberá et al. (2021) suggest that even dictionaries designed for specific tasks might not perform as well as supervised learning methods. This is because the dictionary approach restricts keywords to a specific range and ignores the use of synonyms. In this study, the data to be processed comes from different creators on YouTube. They have different language styles, so setting up an exhaustive dictionary is very difficult.

The third approach is human rating. This method uses multiple evaluators to score the strength of the text's political views. However, this approach is largely interfered with by subjective judgment. Due to differences in personal experience and knowledge, multiple evaluators may give conflicting results, especially with regard to the diversity of expression on YouTube. Variations in personal experiences and knowledge among evaluators can lead to disparate results. This issue is particularly pronounced given the diverse modes of expression found on platforms like YouTube, where the range of content styles can influence the interpretation of political viewpoints.

To overcome these limitations, this study introduces an innovative use of a pre-trained transformer language model to measure the differences in the strength of political views within video content. The Manifestoberta model (Burst et al. 2023) is based on XLM-RoBERTa models (Liu et al. 2019) and fine-tuned on over 1.5 million annotated statements from the Manifesto Corpus. The Manifestoberta model facilitates corpus-based classification across 56 distinct political topics, distinguishing it as the sole language model to date trained exclusively on formal political texts. For any given input text, the Manifestoberta model assigns a predictive score to each of the 56 political topics delineated in the Manifesto Project's coding scheme. These scores indicate the likelihood that the text pertains to specific topics, including social justice. The topic of social justice and the need for fair treatment of all people, as defined by the coding scheme, includes aspects such as special protection for underprivileged social groups, removal of class barriers, the need for fair distribution of resources, and the end of discrimination, whether racial, sexual, or otherwise. In other words, if a video contains more content that aligns with the topic of social justice, then the model gives a higher prediction score. The prediction score accurately measures the strength and focus of political views expressed in video content.

This chapter explores the correlation between the political content of videos and the political affiliations expressed within the comments. Hypothesis 3 suggests that videos with strong political messages are more likely to receive comments with similar political views. The dependent variable in this analysis is the comments that share the video's political stance. Two solutions are considered to quantify this: one method counts the number of comments per political category and creates a count model, and the other calculates the percentage of each political category in the comments and creates a linear model. The reliability of comment counts as a measure is uncertain because each video has a different likelihood of being viewed and commented on, influenced by the recommendation algorithm. Including view counts and comment counts as control variables introduces a risk of multicollinearity, as these factors may be highly correlated with each other. A linear regression analysis was conducted to quantitatively analyze the relationship. The strength of the video's political content on social justice topics was the independent variable, and the proportion of L-label comments for each video was the dependent variable.

Some other factors could impact the dependent variable and thus were included in the regression as control variables. These factors are the year of video publication, the total number of views, and the number of subscribers to the video creator's channel. As shown in Figure 1 and Figure 2, time plays a crucial role in public attention. In some years, the public may be particularly concerned about income inequality due to social events. The total number of views measures the reach and potential impact of the video. The extent of communication may affect the distribution of political positions in video comments, e.g., by being more diverse. Similarly, the number of subscribers could indicate the creator's influence and the likelihood of attracting a broader audience, which might also contribute to a wider range of political opinions in the comments. By including these control variables, the regression analysis can more accurately assess the direct impact of the video's political content on the political affiliations expressed in the comments, ensuring a more robust and reliable result.

In summary, the regression model is defined as follows:

$$L \text{ proportion}_i = \beta_0 + \beta_1 \cdot \text{SocialJusticeScore}_i + \beta_2 \cdot \text{Views}_i + \beta_3 \cdot \text{Suscribers}_i + \beta_4 \cdot \text{Year}_i + \epsilon_i$$

The linear regression model must satisfy three key assumptions: linearity, homoscedasticity, and normal distribution of residuals. An initial check confirms that these assumptions are met. However, the model contains some high-leverage points that interfere with the calculation of the coefficients. Therefore, four points were removed since their leverage value is over three times greater than the

average leverage value. After removing the value of leverage, the new model was found to be heteroscedasticity by the Breusch-Pagan test. Heteroscedasticity occurs when the variance of the residuals is not constant across all levels of the independent variables. While heteroscedasticity does not bias the estimated coefficients, it renders the standard errors invalid. This leads to unreliable hypothesis tests and confidence intervals because the standard errors underestimate or overestimate the true variability of the coefficients. To eliminate the effect of heteroscedasticity on the standard errors, heteroscedasticity-consistent standard errors were used. These robust standard errors correct for the presence of heteroscedasticity, ensuring that the standard errors of the estimated coefficients are reliable.

7.2. Results

Figure 13 shows the distribution of predicted scores on the topic of social justice by the Manifesto model. The violin plot illustrates the density of scores at different levels, with a wider section indicating a higher concentration of data points. A vertical line inside the violin represents the median and the interquartile range of the scores. The shape of a violin is thick at both ends and thin in the middle. This means that on the topic of social justice, most videos are at two extremes. Most of the videos either strongly call for a more equitable distribution of income or not at all. In addition, the higher end is thicker than the lower end, indicating that more videos tend to favor social justice. The prediction scores confirm the finding in the sixth chapter that suggests a few videos are related to the narrative “bad government intervention.”

Figure 14 shows the distribution of L-labeled comments in each video. In contrast, the distribution of comments that support redistribution and against inequality is, in general, low. Most videos have only 10 to 20 percent of L-labeled comments. As discussed in the limitation of the seventh chapter, the fine-tuned BERT model tends to underestimate the L-labeled comments. Thus, there could be more actual L-labeled comments.

Table 7 shows the results from the linear regression. It consists of four models with varying specifications. Model 1 includes only the predicted score as an explanatory variable. Model 2 adds view and subscriber counts as control variables to Model 1. Model 3 includes all variables from Model 2 plus year dummies. The results indicate that in all models, the predicted score has a positive and statistically significant impact on the proportion of L-label comments. This suggests that the stronger the video content is in favor of social justice, the more the comments become in

Figure 13: Distribution of Predicted Score on the Topic Social Justice

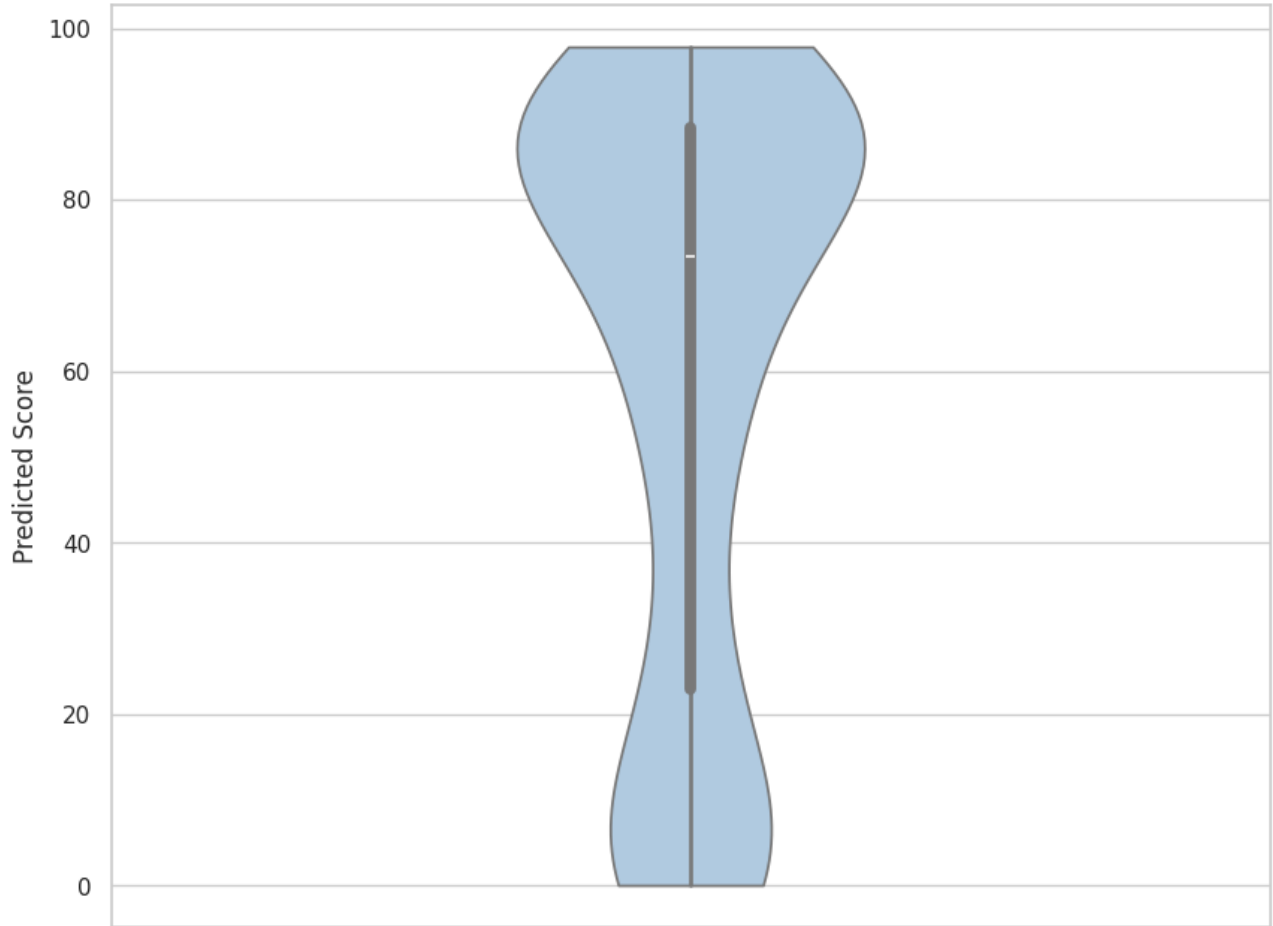


Figure 14: Distribution of L Proportion

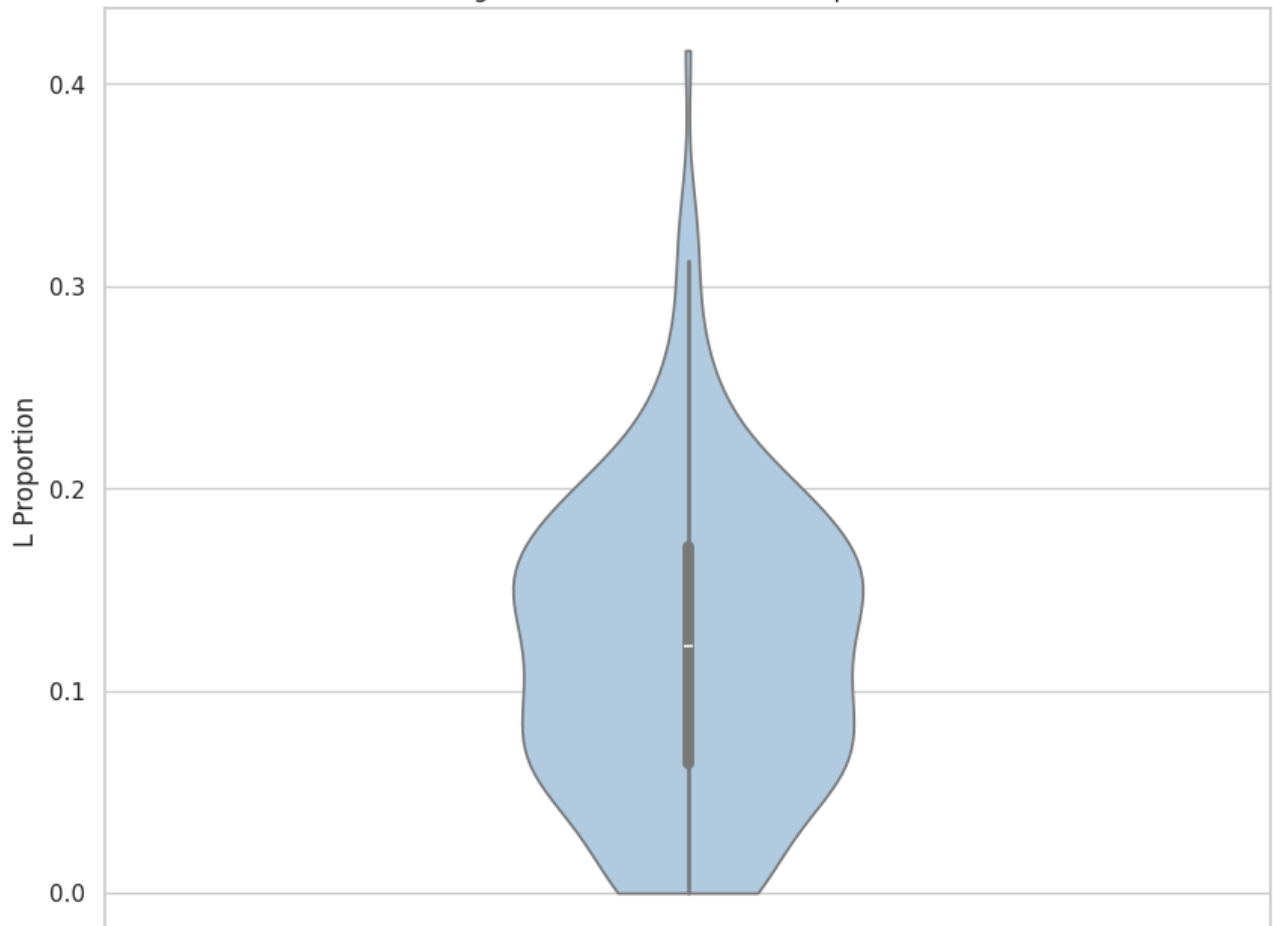


Table 7: Regression Results

	Dependent variable: the proportion of L label comments			
	Model 1	Model 2	Model 3	Model 3 Heteroskedasticity-Robust Standard Error
	(1)	(2)	(3)	(4)
Predicted Score	0,00035*** (0,00013)	0,00035*** (0,00013)	0,00023* (0,00013)	0,00023* (0,00012)
View Count		-0,00000 (0,00000)	-0,00000 (0,00000)	-0,00000 (0,00000)
Subscriber Count		-0,00000** (0,00000)	-0,00000* (0,00000)	-0,00000* (0,00000)
as.factor(year)2008			-0,00910 (0,06420)	-0,00910 (0,10916)
as.factor(year)2009			0,01035 (0,07892)	0,01035 (0,01151)
as.factor(year)2010			-0,05421 (0,06425)	-0,05421 (0,05368)
as.factor(year)2011			-0,05751 (0,05395)	-0,05751** (0,02427)
as.factor(year)2012			-0,06769 (0,05182)	-0,06769** (0,02853)
as.factor(year)2013			-0,01760 (0,04851)	-0,01760 (0,01795)
as.factor(year)2014			-0,04410 (0,04734)	-0,04410*** (0,01528)
as.factor(year)2015			-0,09628* (0,04932)	-0,09628*** (0,02632)
as.factor(year)2016			-0,10738** (0,04900)	-0,10738*** (0,01878)
as.factor(year)2017			-0,05090 (0,05146)	-0,05090* (0,02772)
as.factor(year)2018			-0,07881 (0,04809)	-0,07881*** (0,01647)
as.factor(year)2019			-0,04363 (0,04686)	-0,04363*** (0,01491)
as.factor(year)2020			-0,03355 (0,04821)	-0,03355* (0,01739)
as.factor(year)2021			-0,09884** (0,04763)	-0,09884*** (0,01543)
as.factor(year)2022			-0,09262* (0,04732)	-0,09262*** (0,01550)
as.factor(year)2023			-0,05718 (0,04732)	-0,05718*** (0,01675)
as.factor(year)2024			-0,09666** (0,04751)	-0,09666*** (0,01747)
Constant	0,09729*** (0,00881)	0,10309*** (0,00919)	0,17525*** (0,04608)	0,17525*** (0,01177)
Observations	239	239	239	
R ²	0,03111	0,04868	0,21518	
Adjusted R ²	0,02702	0,03653	0,14318	
Residual Std. Error	0,06833 (df = 237)	0,06800 (df = 235)	0,06412 (df = 218)	
F Statistic	7,60873*** (df = 1; 237)	4,00800*** (df = 3; 235)	2,98854*** (df = 20; 218)	

Note:

* p<0,1; ** p<0,05; *** p<0,01

favor of redistribution and against inequality. The coefficient for the predicted score is 0.00023. This means that for every one-unit increase in the predicted score, the proportion of L label comments is expected to increase by 0.023%, assuming all other variables in model 3 remain constant. The subscriber count has a minimal and significant negative impact on the proportion of L-label comments. It is worth noting that the impact of the year dummies in model 3. Compared with the year 2007, the coefficients for other years are negative and, in some years, are highly significant. This likely hints at the significant impact of social events that occurred that year on social media comments.

The results of linear regressions confirm hypothesis 3. The results of the regression show that as the association of video content with social justice rises, there are more comments that tend to be against inequality and in favor of redistribution. This finding confirms Mukerjee and Yang's conclusion (2021) that no matter how popular news is on social media, users typically prefer news from sources they agree with and Cinelli's argument (2020) that people tend to be exposed to news with consistent positions. However, this effect of attracting like-minded comments is very weak and smaller than year dummy variables.

There are two possible reasons for the weak echo chamber effect. First, YouTube's distribution model does not rely on following networks since the results show the number of subscribers has a negligible impact on the distribution of political positions in comments. Instead of following specific accounts, users typically use the search function to find videos on topics they are interested in. This leads to a diverse range of content appearing in search results, increasing the likelihood of users being exposed to varied viewpoints. Second, the population living in echo chambers is very small. Research by Nelson et al. (2017) and Flaxman et al. (2014) indicates that most people consume diverse political news. Preferences for news with a particular political stance are more closely related to personal experiences (Hameleers et al. 2018).

This finding fills a research gap on how variations in political affiliation intensity affect social media consumption patterns. Building on this, future research could further explore the impact of algorithmic recommendations on user comments. While YouTube's distribution model doesn't heavily depend on follower networks, its recommendation algorithm significantly influences which videos users are likely to see. Experimental methods could be employed in future studies to examine how these algorithms affect user engagement with diverse viewpoints or reinforce existing

beliefs. For instance, researchers could analyze changes in user behavior after interacting with recommended videos compared to those they actively search for.

7.3. Limitations

The accuracy of the results hinges on two key variables derived from the fine-tuned BERT model and the Manifestoberta model. As noted in the seventh chapter, the fine-tuned BERT model tends to underestimate the proportion of L-labeled comments, which could lead to inaccuracies in the regression analysis. Furthermore, the predictive scores related to social justice are influenced by the diversity of the corpus used to pre-train the model. If this corpus lacks a broad range of viewpoints, the model may develop biases, which could skew the prediction scores toward certain perspectives or narratives.

While the regression models attempt to control for various factors like view and subscriber counts, as well as year dummies, other variables with impact may not be accounted for. These unaccounted variables can introduce additional noise and bias to the results.

8. Conclusion

This research investigates the unique characteristics of YouTube as a major online video platform and its impact on public discourse regarding income inequality. The study focuses on three dimensions: content production, distribution, and consumption. Firstly, the research explores how the discussion on income inequality is divided on YouTube. This involves examining the degree of narrative fragmentation within these discussions, considering the platform's decentralized nature that allows diverse and sometimes opposing narratives to coexist. Secondly, the study investigates the polarization of interaction networks in videos about income inequality. This question delves into the extent of group polarization within YouTube's interaction networks, particularly in the comments sections of videos discussing income inequality. It aims to determine whether these interaction patterns reflect the homogenization observed on other social media platforms such as Facebook and Twitter. Lastly, the research examines to what extent video content attracts comments of the same political stance. This involves analyzing how the ideological orientation of video content influences the homogeneity of the comments it attracts.

Using BERTopic, the study found that narrative divergence occurs in income inequality videos. Four distinct topics emerged from the analysis: Mainstream Narrative, Wealth Accumulation, Broken Capitalist System, and Bad Government Intervention. The Mainstream Narrative was identified as the context topic representing a comprehensive discussion on income inequality and covering various perspectives. The Wealth Accumulation topic focuses on the rapid concentration of wealth among a few individuals, its impact on political influence, and the need for policies that distribute wealth more equitably. The Broken Capitalist System topic offers a critical perspective on capitalism's emphasis on maximizing short-term shareholder profits at the expense of workers and societal welfare. In contrast, the Bad Government Intervention topic advocates for minimal government involvement in the economy. It argues that excessive regulation and high taxes discourage economic competition and personal initiative, potentially leading to negative outcomes such as poverty traps. The last three topics are clearly separated from the mainstream narrative and independent of the others. Moreover, the Broken Capitalist System and the Bad Government Intervention appear to have completely opposite understandings and claims about income inequality. While 71% of videos were classified under the mainstream narrative, 28.5% focused on singular viewpoints, raising concerns about the potential for a skewed or incomplete understanding of income inequality.

The analysis of YouTube comments on videos related to income inequality provides insights into user interaction patterns and explores the presence of political polarization on the platform. The results indicate that most comments (71.8%) do not clearly express political attitudes, suggesting YouTube is not predominantly used for political debates. Among the comments that identified a political position, 11.9% are left-leaning, which is against inequality and supports redistribution, and 16.4% are right-leaning, which justifies inequality and is against government intervention. The examination of the E-I index reveals that both left-leaning and right-leaning comments show higher engagement with external links than internal ones, resulting in positive E-I indices (0.38 for left-leaning and 0.41 for right-leaning comments). This suggests that users with political stances on YouTube tend to argue more with those holding opposing views rather than forming homogeneous groups. Despite the general trend of heterogeneous interactions, outliers in the boxplot for right-leaning comments and the long downward whisker for left-leaning comments indicate that some videos exhibit homogeneous interaction patterns. This is particularly evident in videos with fewer comments, where individual interactions can disproportionately influence the overall pattern, leading to internal focus and reduced exposure to opposing viewpoints. Overall, the positive global E-I index and the findings from multiple videos suggest that YouTube users predominantly engage in discussions with individuals holding different political affiliations. This trend challenges the notion that social media platforms inherently foster political polarization and highlights the complexity of interaction patterns on YouTube. Future research should continue to explore how platform design influences user interactions and the potential for cross-partisan communication to mitigate polarization.

In the final part of the empirical analysis, I used a combination of deep learning and statistical regression to investigate the impact of video content on viewer comments. The distribution of predicted scores indicates significant polarization, with most videos either strongly supporting or opposing equitable income distribution. Notably, more videos lean towards pro-social justice views, supporting earlier findings of narrative divergence on YouTube. The linear regression analysis reveals that videos with stronger social justice content tend to attract more comments favoring redistribution and opposing inequality. Specifically, for every one-unit increase in the predicted social justice score, the proportion of left-leaning comments increases by 0.023%. This suggests that the content's ideological orientation significantly influences the type of engagement it receives, supporting the hypothesis that users prefer content aligning with their beliefs. Interestingly, the

overall effect of this alignment is relatively weak. The minimal impact of subscriber counts and the significant influence of yearly variations suggest that broader social events and YouTube's search-based distribution model, which exposes users to diverse viewpoints, play crucial roles in shaping interaction patterns. These findings provide a nuanced understanding of YouTube as a platform for political discourse. While content can attract like-minded comments, the weak echo chamber effect and the diverse exposure provided by the platform suggest a complex interaction environment. Future research should further investigate the role of recommendation algorithms and explore how different platform characteristics influence user engagement and political polarization.

The results could be affected by several methodological limitations. Firstly, the BERTopic model struggles to distinguish between topics with high semantic similarity. This means some potential topics might not be identified due to their extreme similarity. Prediction errors in the fine-tuned BERT model also pose a limitation. The model underestimates the counts for specific labels (R and L), often misclassifying them as N. This misclassification can lead to inaccuracies in the E-I index by overestimating external links. Additionally, the study's reliance on the fine-tuned BERT model and the Manifestoberta model introduces further complexities. The underestimation of L-labeled comments and the influence of the training corpus's diversity on predictive scores may skew the results. Furthermore, unaccounted variables in the regression models could introduce noise and bias, affecting the reliability of the findings. Nevertheless, these biases are not sufficient to overturn the findings reached in this study.

The findings contribute to academic research in two main ways. Firstly, this study identifies narrative divergence on YouTube, filling a gap in the analysis of social media content. Future research can explore how different narratives on income inequality affect users' attitudes. Secondly, the study finds that group polarization seen on Facebook and Twitter does not occur on YouTube. This shows that platform-specific features influence user interactions. The design and algorithms of social media platforms play a crucial role in shaping discourse. Future research should examine these platform-specific characteristics more closely, looking at how different social media environments affect the formation of echo chambers and polarization. Additionally, advanced natural language processing techniques, like BERTopic and fine-tuned BERT models, prove effective in analyzing complex social media data. However, the study also points out methodological limitations, such as difficulty distinguishing similar topics and prediction errors. Future research could improve these methods by using models specifically trained on political texts,

better text filtering techniques, and larger, more balanced training datasets. Furthermore, future studies should explore how recommendation algorithms shape user exposure to different viewpoints. Understanding how these algorithms influence content visibility can show whether they reduce or increase polarization. Experimental studies could help assess how algorithm changes impact user behavior and engagement.

In summary, this study enhances the understanding of YouTube's role in shaping public discourse on income inequality, highlighting both its potential to foster diverse discussions and the complexities of its interaction environment. By addressing the identified limitations and expanding the scope of future research, scholars and practitioners can continue to improve the design and regulation of social media platforms to support healthy public discourse.

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10. Appendix

Figure A: Term Score decline per Topic

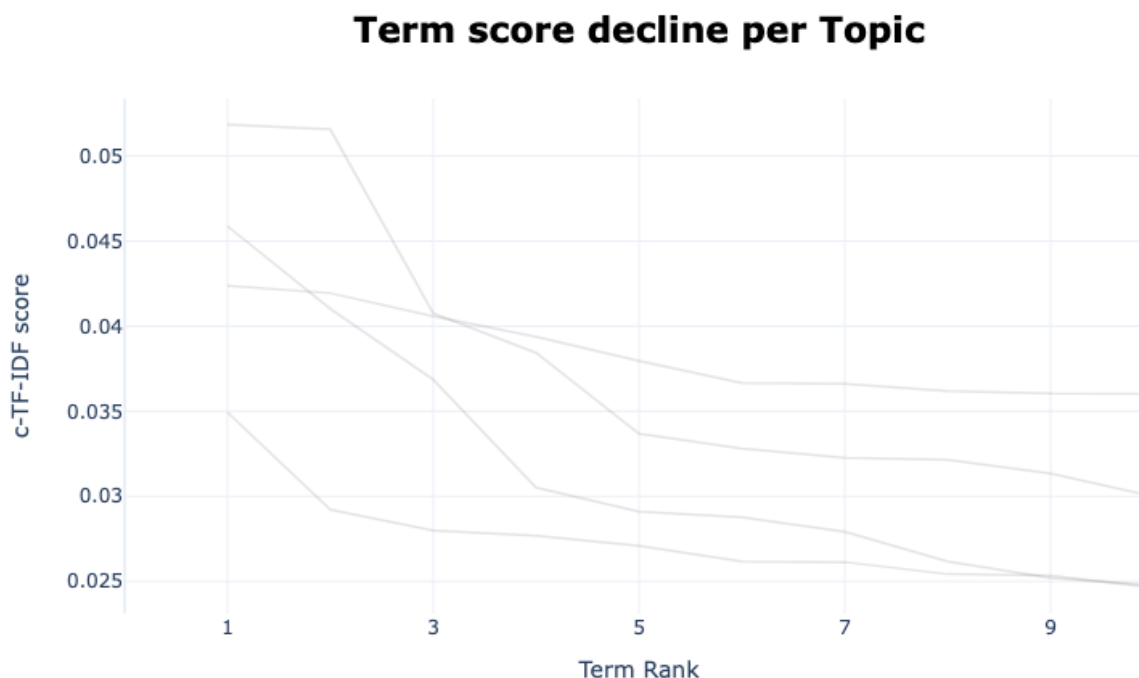


Figure A shows the decline in the term score per topic. The original graph is an interactive line chart drawn by Plotly. The BERTopic's creator fixes the function's parameters. The y-axis indicates the TF-IDF score. The x-axis represents the ordinal number of the word represented by each topic. The lower the word ranking, the less unique the word is about the topic it belongs to.

Figure B shows label distribution in primary comments and secondary replies.

Figure C shows the linearity between predicted scores and L proportions in comments.

Figure D compares the distribution of studentized residuals of Model 3 and the normal distribution.

Figure E shows the scatterplot of residuals and subscriber count. The spread of residuals seems larger at higher subscriber counts, which could lead to heteroscedasticity (increasing variance of residuals with increasing values of the independent variable).

Table A indicates the variance inflation factors in Model 3. All values are below 1.5, and therefore no multi-linearity in model 3.

Figure B: Label Distribution by Reply Status

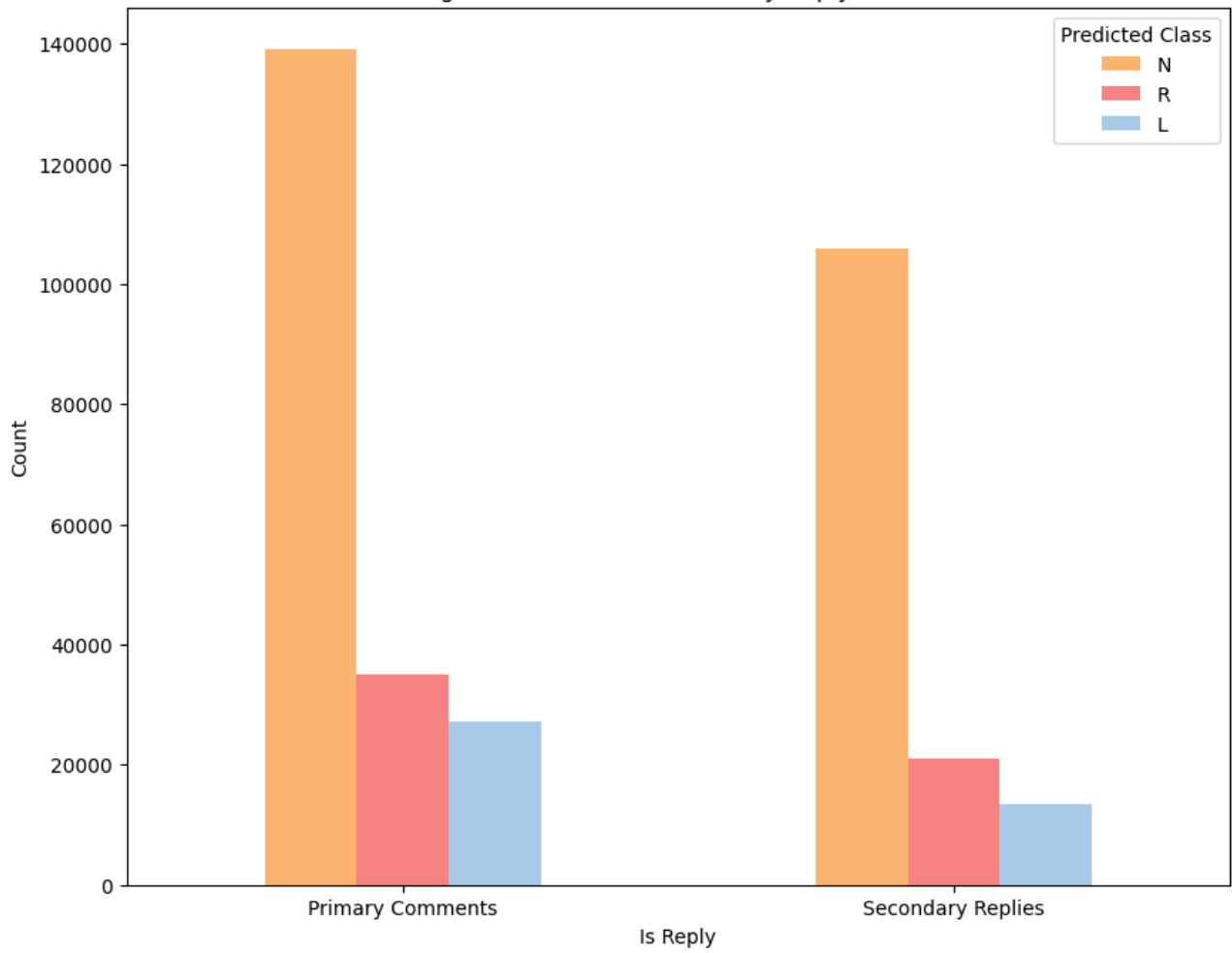


Figure C: Relationship Between Predicted Score and L Proportion

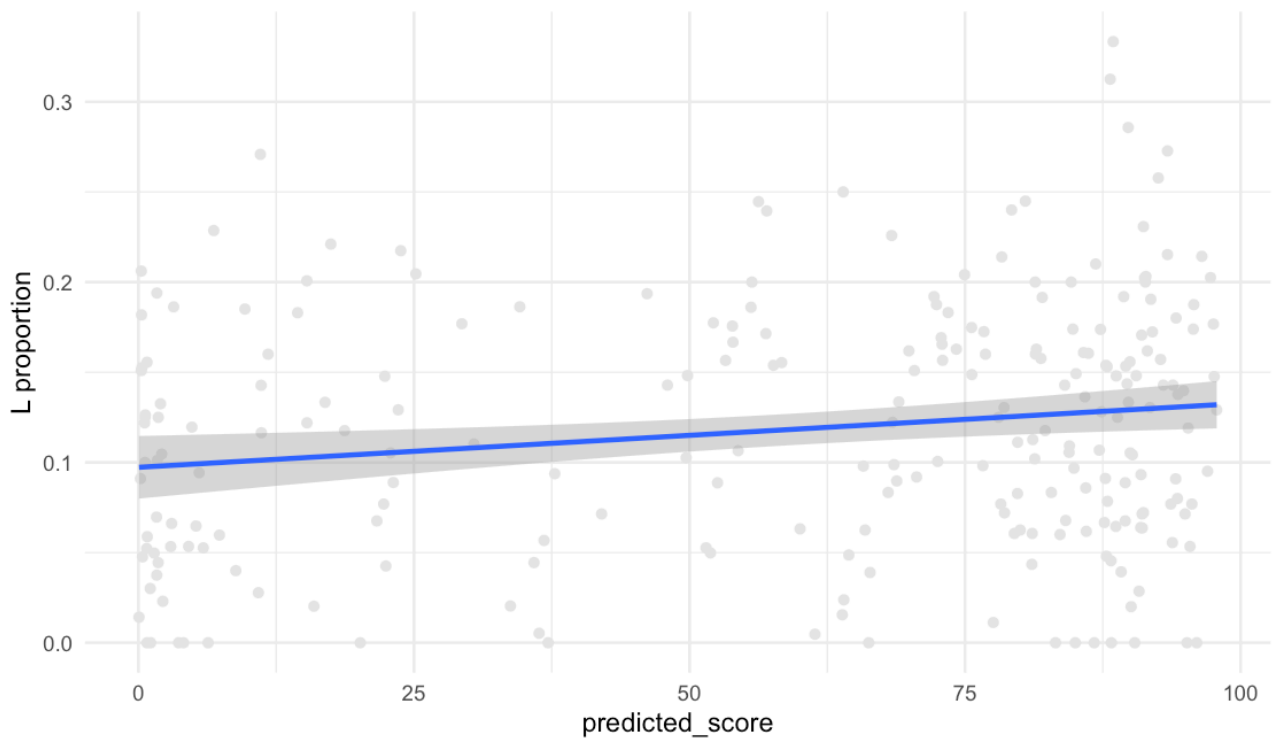


Figure D: studentized residuals vs. normal distribution

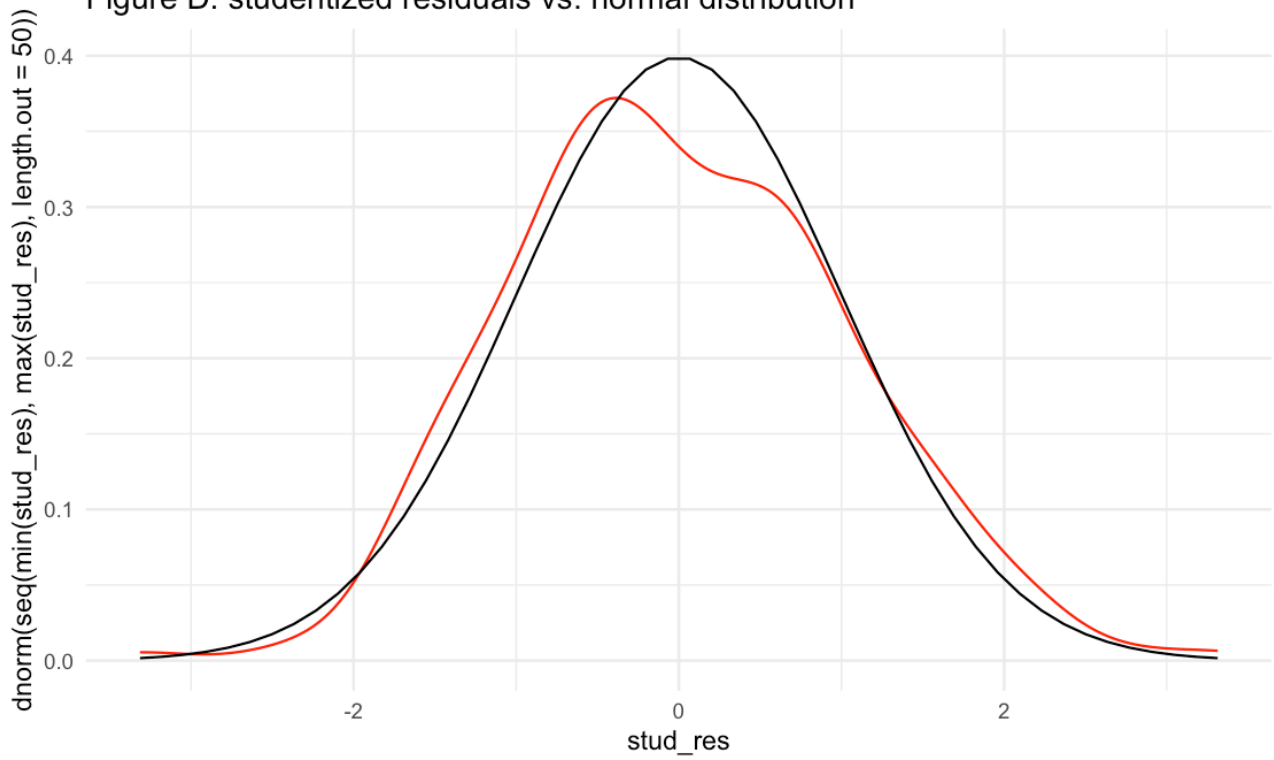
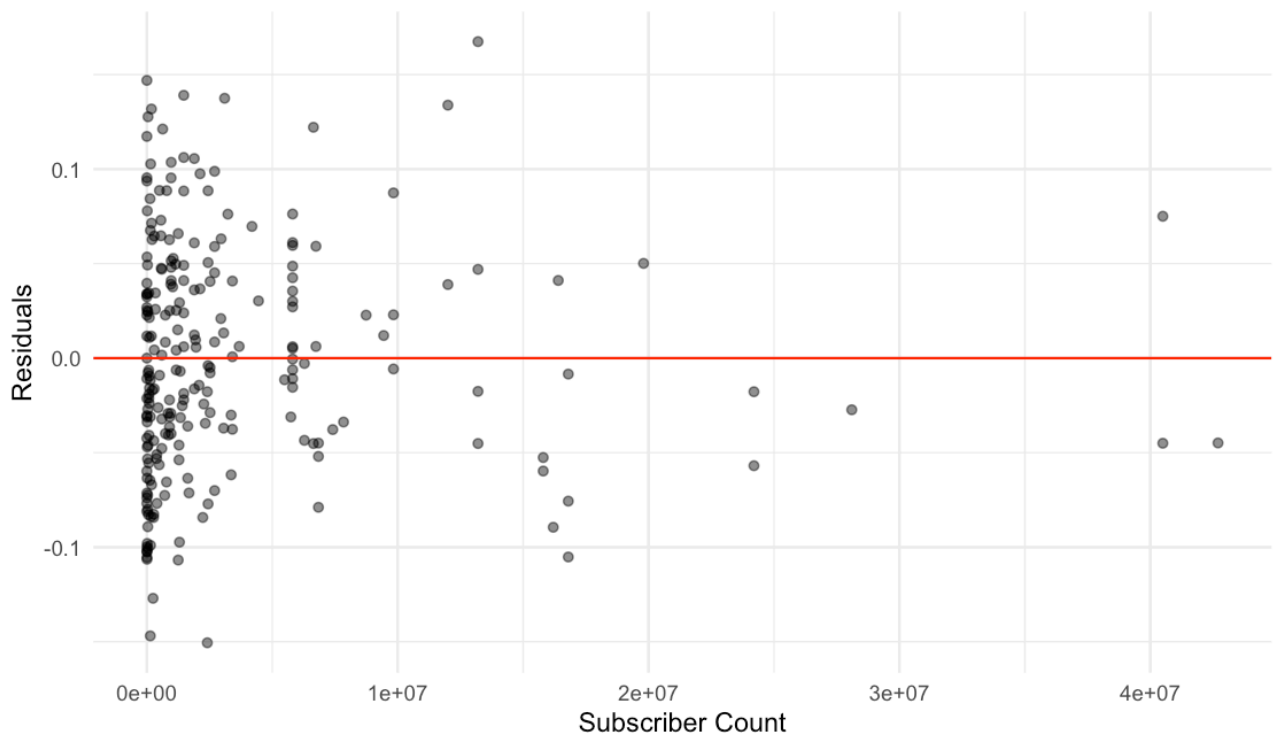


Figure E: Residuals vs Subscriber Count



	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
predicted_score	1.120127	1	1.058361
view_count	1.079057	1	1.038777
channel_subscriber_count	1.057360	1	1.028280
as.factor(year)	1.264597	17	1.006928

Table A: Variance inflation factors in Model 3

11. Supplemental materials

Code script	Purpose	Generated documents	Generated graphics
1-MasterThesis-Videos	Collecting videos and their statistics	video_index_edited.csv	
2-MasterThesis_Comments	Collecting videos' comments	Comments	
3-MasterThesis_Data_description	Creating data description	all_comments.csv	Figure 1, 2, 3, 4, 5
4-MasterThesis_BERTopic	Filtering text and modeling topics	filtered_data.csv topics_data.csv	Figure 6, 7, 8, 9, A Table 1, 2
5-MasterThesis_Shuff	Random sampling training set	train_set.xlsx train_set05.xlsx	Table 3 Figure 10
6-MasterThesis_FineBERT	Fine-tuning BERT model and predicting labels of comments	BERTmodel all_comments_with_labels.csv	Figure 11, 12, B Table 4, 5, 6
7-MasterThesis_Manifesto	Measuring viewpoints and creating a document for R	Videos_index_forR.csv	Figure 13, 14
8-MasterThesis-Regression	Running linear regression		Table 7, A Figure C, D, E

The supplemental materials are saved on [One Drive](#).

1. The Raw Data folder stores the raw data crawled through the API, and the Refine Data folder stores the processed files that are passed on to the next step. The Outputs folder stores the results from chapter 5 and chapter 6.
2. The train_set05.xlsx is the manually labeled training set.
3. The BERTmodel folder stores the fine-tuned BERT model, including tokenizer and vocabs.
4. The fine-tuning BERT model script is run on an A100 40GB, and the fine-tuning process requires about 34GB of graphics memory.
5. The first seven scripts are all Python (3.11) programming. The last regression script is an R (4.3.1) markdown document.

12. Declaration of Using AI-Tool

This thesis used Chat-GPT to improve the clarity, coherence, and simplicity of the text. AI modified the text based on my original idea and content. AI-generated text is selectively applied to the paper.



Declaration of Originality

I hereby declare that the qualifying thesis has been written by myself without unauthorized assistance and that I have observed the principles of academic honesty.

(cf. https://api.swiss-academies.ch/site/assets/files/25607/kodex_layout_en_web-1.pdf)

01.06.2024, Zurich

.....
Date and place

Signature